Classifying Market Regimes

What to measure, which regimes to name, where training labels come from, and what a causal classifier can learn

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Abstract

Classifying the market regime usually means forecasting part of the future. The regimes worth building are defined by how much the market will move, which can be forecast. Regimes defined by which way it will move cannot be forecast, according to a century of US data.

A regime definition is a choice. Eight reasonable definitions of a bad market, from bear markets to recessions to volatile months ahead, agree only moderately over a century of US data. Most regime vocabularies divide five quantities into a few states: risk, the compensation for risk, drift, the economy, and co-movement. Many labels use future data. A day counts as part of a bear market because prices fell 20% afterward. From 1927 to 2025, leaving the market during bear markets known in hindsight would have raised the Sharpe ratio from 0.45 to 1.54. Leaving when the 20% rule actually confirmed each bear market gave 0.46. The confirmation came a median of 70 trading days after the peak.

Risk persists, so it can be forecast. A simple model ranked a day before a volatile month above a day before a calm month 88% of the time, and last month's volatility alone did so 84% of the time. The same model forecast the sign of the next quarter's return slightly worse than a coin flip. The VIX, credit spreads, the yield curve and the volatility term structure largely repeat what recent price variation already shows. In the worked example they added almost nothing to seven price-based features, and 19 more years of history were worth more than any extra feed. Each label source has a reach: how far into the future it looks. The reach sets how the training data must be split and how late any detection will be. Official recession dates arrive 7 to 15 months late on average.

Because regimes persist, the most recent known label is a strong baseline, and a model must beat it, not a coin flip. A small, regularized model usually matches a complex one, because there are only a few dozen crises to learn from. A quarter-century of daily data holds nine bear markets, so evaluate per episode, with error bars that reflect the small count. The chapter recommends defining the regime by the risk a decision depends on over the next few weeks, and never training on a label before it could have been known.

What this is. A practitioner who sets out to "detect the market regime" meets four questions that the phrase hides. Which definition of a regime should the study adopt? Which observable inputs carry information about it? If a model is to be trained, where do the training labels come from? And which model turns the inputs into a classification? This chapter answers them in that order (§2–§5). Its companion, Market Regimes and Hidden Markov Models, treats a regime as a latent state and builds the hidden Markov model that infers it. This chapter treats a regime as a *target*: something a decision needs, defined explicitly, and estimated from data available at the time. Most of its length goes to the two questions the latent-state view leaves open: what to feed a classifier, and what to train it against.

How to read it. Part I (§1–§2) states the problem and surveys the regime definitions practitioners use. Part II (§3–§4) covers inputs and labels, the two sections a reader building a regime study will use most. Part III (§5–§8) covers classifiers, a worked example on public data from 1926 to 2025, evaluation, and the architecture of a production study. §9 is the synthesis. If you read three things, read §1.4 (the reach of a label, and why detection is forecasting), §4.9 (vetting a label before training on it) and §6 (what a causal classifier learns on public data, with labels over a century and classifiers since 1990). Appendix A defines the statistical and financial concepts the main text uses without deriving them.

Relationship to the other notes. Market Regimes and Hidden Markov Models supplies three results used throughout: regime models identify volatility and not expected returns; the choice between predicted, filtered and smoothed state estimates governs measured performance; and the effective sample size of any regime study is its number of episodes, not its number of days. It is the prerequisite for §5.7 and §5.8. Implied Volatility derives the VIX and the variance risk premium used in §3.3. Bonds and Bond Markets covers the credit spreads and yield curves of §3.4 and §3.5. Trend-Following in Financial Markets covers the trend estimation behind §2.4. Edge as Information relates hit rates to information, which §1.4 uses to size what a good direction signal delivers. Each note stands alone.

Epistemic tags. Claims are flagged by status where the status changes what you should do:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement among people who have looked.

- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. It may well be right; the evidence is private or absent, which is not the same as evidence against it.

Untagged sentences are definitions, derivations, or arithmetic. The numbers in §6 and every number attributed to "the worked example" come from public data and are printed by `figures/rc_common.py`, so you can change a parameter and rerun.

Notation. S_t is the level of a total-return price index at the close of trading day t, $r_t = S_t/S_{t-1} - 1$ its simple return and $\ell_t = \log(1 + r_t)$ its log return. $A = 252$ is the number of trading days per year. $\mathcal{F}_t$ is the information available at the close of day t: data *published* by then, not data that describe day t but appear later.

z_t is the vector of latent **state variables** (§1.3, §2.2): the quantity of risk, the price of risk, the direction of the market, the macroeconomic state and the structure of co-movement. $x_t \in \mathbb{R}^d$ is the vector of **inputs** (features) known at t; they relate to the state through $x_t = \phi(z_t) + \varepsilon_t$, with ε_t the measurement noise (§1.3). A **regime** $s_t \in \{1, \dots, K\}$ is a coarse partition of the state, $s_t = c(z_t)$, into K classes (coded $\{0, 1\}$ when $K = 2$, as for a label). Lowercase s_t is the regime and uppercase S_t the price; the two never appear in the same role. As in the companion chapter, vectors such as x_t and z_t are set in plain type; Σ denotes a covariance matrix.

A **label** y_t is the training target attached to day t, computed from the price path (or from other data) by a rule L: $y_t = L(S_{t-a}, \dots, S_t, \dots, S_{t+b})$. Its **lookback** is a and its **reach** is b, the number of days after t on which it depends (§1.4). For a label defined over a forward window of fixed length, $b = h$, the horizon. Labels are binary unless stated, with $y_t = 1$ marking the class of interest (the bear market, the high-volatility month). $p_t = \Pr(y_t = 1 \mid \mathcal{F}_t)$ is the best possible probability forecast of the label, and $\widehat{p}_t$ a classifier's estimate of it. An **episode** is a maximal run of consecutive days with $y_t = 1$; a label's episode count is the number of such runs, and an episode's length the number of days in it.

$\mathrm{RV}_t^{(n)} = \sqrt{(A/n)\sum_{i=0}^{n-1} \ell_{t-i}^2}$ is the trailing n-day realised volatility, annualised. VIX_t is the Cboe volatility index in percentage points. δ is the threshold of a turning-point rule (a 20% move unless stated), and τ denotes particular dates: a peak, a trough, a confirmation, a refit. $\Lambda(u) = 1/(1 + e^{-u})$ is the logistic function; σ is reserved for volatility and never means a sigmoid. β collects the coefficients of a classifier and α the ridge penalty on them (§5.4), λ is the switching penalty of a jump model (§4.6), and κ is Cohen's kappa (§6.2). AUC, the Brier score BS and the Brier skill score BSS are defined in §7.1–§7.3 (informally in §1.4 and §5.1). In §5.11, p_{jk} with two class subscripts is a Markov transition probability, as in the companion chapter; p_t and $\widehat{p}_t$, with a time subscript, are label probabilities. A symbol used in one subsection only is defined there.

What you will be able to do. After this chapter you should be able to:

1. Turn "detect the regime" into a stated decision, a state variable, a label and a horizon, and say in advance how much of the label a classifier can learn.
2. Choose inputs for each state variable, align them to the time they were published, and transform them so that a model trained on one decade applies to the next.
3. Construct training labels from six kinds of source, and measure each label's reach, persistence, stability and episode count before training anything.
4. Build a walk-forward training set that never uses a label before it was knowable, and score a classifier against the baseline that matters.
5. Choose among threshold rules, composite indices, logistic models, tree ensembles, latent-state models and teacher–student pipelines for a given decision, and know how each one fails.

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- A. Concepts and prerequisites

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1. What classifying a regime means

1.1 The wrong intuition

The intuition most people bring to the problem goes like this. At any moment the market *is* in some regime: a bull market or a bear market, risk-on or risk-off, calm or stressed. The regime is a fact about the world, like the state of a machine. A good enough model with enough inputs reveals it, and the history of past regimes supplies the training labels. Under this view regime detection is fault detection: more sensors give earlier and surer detection.

The intuition fails in three places.

1. **A regime is a definition, not a fact.** Eight reasonable definitions of a "bad" regime on the same century of US data agree only moderately with one another. Their pairwise Cohen's kappa, a chance-corrected agreement score with 1 for perfect agreement and 0 for agreement no better than chance, ranges from 0.03 to 0.58 for definitions in different families (§6.2). There is no single regime history to recover.
2. **The definitions practitioners care about depend on the future.** A day is in a bear market, in the usual sense, only if prices go on to fall 20% from the peak before it. A classifier asked to say so before that fall is complete is being asked to forecast it.
3. **The parts of a regime that are knowable now need measurement, not classification.** Today's volatility, drawdown and credit spread are observable. A rule computes them; nothing has to be learned.

One number makes the second point concrete. Over 1927–2025, a strategy that held the US market when the ±20% bull–bear label said "bull" and cash otherwise earned a Sharpe ratio of 1.54, against 0.45 for holding the market throughout. That uses the label one day ahead. Acting on the label only once the rule itself has confirmed it, the Sharpe ratio was 0.46 (§6.3). In Sharpe-ratio terms, the label's entire value sits in the days when it cannot yet be known.

The right question is therefore not "which regime are we in?" It is "which property of the coming weeks or months matters to my decision, how well can it be forecast from what I see now, and is a discrete regime the best way to carry that forecast?"

1.2 Three questions before any data

A regime study should begin with a sentence, not a dataset. The sentence answers three questions.

1. **Regime of what?** The behaviour of which asset, portfolio or strategy is being classified. A regime of the S&P 500's volatility, a regime of a momentum strategy's returns and a regime of the stock–bond correlation are different objects with different inputs.
2. **For which decision?** Sizing, hedging, choosing between strategies, setting risk limits, or conditioning another model. The decision says which outcome the label must separate.
3. **At what horizon?** The decision's holding period sets the label's horizon. A daily volatility-targeting rule needs next-week or next-month risk; a strategic allocation needs the next year's macroeconomic state.

Written out, a well-posed study reads like this: *"Reduce equity exposure when the probability that realised volatility over the next 21 trading days lands in the top fifth of its history exceeds 0.6."* Every term in that sentence is measurable. "Detect the regime" is not.

The decisions fall into three groups, and the groups differ in how much a model can deliver:

Decision	Outcome the label must separate	Learnable from current data?
Risk: sizing, hedging, limits	future volatility, drawdown, correlation	largely — risk is persistent (§6.4)
Direction: timing, trend exposure	future return, bull or bear phase	barely — direction is not identified (§6.4)
Macro allocation: duration, inflation hedges, factor tilts	growth, inflation, recession	partly — slowly, with lags and revisions (§2.8)

The rest of the chapter elaborates this table. Risk-type regimes are where regime studies earn their keep. Direction-type regimes are where most of them are aimed.

1.3 The master form

Every regime study, whether a VIX threshold or a gradient-boosted classifier trained on jump-model labels, fills the same five slots. In words: a few hidden state variables drive the market; observable inputs measure them with noise; a regime coarsens them into a few classes; a label approximates that coarsening using the realised price path; and a classifier estimates the label from inputs available at the time.

$$
\begin{aligned}
x_t &= \phi(z_t) + \varepsilon_t && \text{(inputs observe the state, with noise)} \\
s_t &= c(z_t) && \text{(a regime is a coarse partition of the state)} \\
y_t &= L(S_{t-a}, \dots, S_t, \dots, S_{t+b}) && \text{(a label is computed from the path)} \\
\widehat{p}_t &= f(x_t, x_{t-1}, \dots) \approx \Pr\left(y_t = 1 \mid \mathcal{F}_t\right) && \text{(a classifier estimates it causally)}
\end{aligned}
$$

The state z_t is continuous and unobserved. §2.2 argues that five state variables underlie every practitioner taxonomy: the **quantity of risk** (volatility), the **price of risk** (risk premia and risk appetite), the **direction** of the market (its conditional mean), the **macroeconomic state** (growth, inflation and policy), and the **structure** of co-movement (correlation, dispersion and liquidity). The function ϕ says how inputs relate to these, and ε_t is measurement noise: the VIX measures the quantity of risk plus a time-varying premium, a credit spread measures expected default plus the price of risk.

The regime $s_t = c(z_t)$ is a discretisation. Its value lies in carrying a discrete decision (on or off, strategy A or strategy B) and in communicating; its cost is the information discarded by the coarsening. The label y_t is the computable stand-in for the regime, the thing a supervised model is trained against. The classifier f is restricted to inputs dated t or earlier.

Five slots vary between studies: the state variable, the coarsening c, the label rule L (including its reach b), the inputs ϕ, and the classifier f. Most named methods are a choice of three of them with the other two left implicit. A VIX-band study chooses the quantity of risk, a coarsening into bands, and a label with zero reach, so it needs no classifier. A bull–bear study chooses direction and a turning-point label, and then usually forgets that the label's reach is months long. The comparison tables of §2.12, §4.8 and §5.12 are this master form, filled in.

The companion chapter's master form describes the same objects from the other side. There the regime is a latent Markov state, the "label" is its posterior probability, and nothing is trained against an external target. The two views meet in §5.7 and §5.8: a latent-state model can be the classifier, or it can produce the labels a supervised classifier then learns.

1.4 The reach of a label, and why detection is forecasting

The **reach** b of a label is the number of days after t that the label of day t depends on. It is the single property that decides what a classifier can learn, how the training set must be built, and how late any detection will be.

First, suppose $b = 0$. Then y_t is a function of data available at t, and the best possible forecast is the rule itself: $p_t \in \{0, 1\}$, and its error is zero. Training a model to reproduce such a label is, at best, a slow way to compute it. "VIX above 30" and "10% below the high" are labels of this kind. They are useful regimes, but they are measured, not learned.

Now suppose $b > 0$. The label depends on what happens next, and the best possible forecast is the conditional probability $p_t = \Pr(y_t = 1 \mid \mathcal{F}_t)$. Even this ideal forecaster has an irreducible error. Measured by the Brier score, the mean squared difference between forecast probability and outcome, the ideal forecaster's error is the expected conditional variance of the label:

$$\mathrm{BS}^* = \mathbb{E}\big[(p_t - y_t)^2\big] = \mathbb{E}\big[p_t\,(1 - p_t)\big].$$

The second equality is the Bernoulli variance: given $\mathcal{F}_t$, the squared error is $(1-p_t)^2$ with probability p_t and p_t^2 otherwise, so its mean is $p_t(1-p_t)$. Hence BS^* is zero only when $\mathcal{F}_t$ settles the label, and it is largest when p_t sits near one half. Whatever a model achieves, in expectation it cannot do better than BS^*, and how large BS^* is depends on what drives the label's future part.

This yields the chapter's first organising claim. **A supervised regime "detector" is a forecaster of its label's future part, and what it can learn is set by what that future part depends on:**

- **If the future part is volatility**, the label is highly learnable, because volatility is persistent. The log VIX has a daily autocorrelation of 0.98 since 1990, a half-life of about 35 trading days (a deviation with daily autocorrelation ρ halves in $\ln 0.5 / \ln \rho$ days). Log 21-day realised volatility on the US market since 1930, regressed on its own value a month earlier, has a slope of 0.72, a half-life of about two months. In §6.4 a label defined by next month's volatility is forecast with an AUC of 0.88. (AUC is the probability that the classifier ranks a randomly chosen positive day above a randomly chosen negative one; 0.5 is chance.)
- **If the future part is direction**, the label is barely learnable. A correlation (an information coefficient) of 0.05 between a forecast and the next period's return, which would be a good equity-timing signal, corresponds to a hit rate of 51.6% (Edge as Information). In §6.4, a label defined by the sign of the next quarter's return scores an AUC below 0.5 out of sample.
- **If the future part is the macroeconomic state**, the label is learnable at low frequency from slow inputs such as the yield curve and credit spreads, with long publication lags and revisions on both the inputs and the labels.

Detection is forecasting. The usual distinction between "detecting the current regime" and "predicting the next one" disappears once reach is in view. Whatever part of today's regime is already settled by today's data can be computed; whatever part is not is a forecast. A study that claims to detect bear markets early is claiming to forecast falls in prices, and must be judged against the evidence on forecasting returns, which is discouraging (Welch & Goyal, 2008).

Random reach. Some labels have no fixed reach. A bull–bear label built from turning points assigns day t to a bear market if t lies between a peak and the following trough, where a peak is confirmed only when the price has fallen $\delta = 20\%$ below it. The label of the days after a peak is therefore settled only on the confirmation date, and its reach is the **confirmation lag**, the number of days from the turning point to its confirmation. On the US market since 1926 the median confirmation lag was 70 trading days for bear markets (interquartile range 42–146) and 41 for bull markets (§6.3). For about a fifth of all days, the label could not have been known until weeks or months later.

The stale-label baseline. Because labels persist, the most recent label that *is* fully known at t is a strong forecast of the current one. For a label with fixed reach b, that is y_{t-b}. For a forward-volatility label with horizon h, y_{t-h} is computed from the returns of $(t-h, t]$: it is trailing realised volatility compared with a threshold that has barely moved in h days, a pure persistence forecast. For a turning-point label it is the **confirmed state**, the rule's own real-time verdict. A classifier must be judged by its improvement over this stale label, not over the base rate. In §6.4 trailing volatility alone reaches an AUC of 0.84 on the forward-volatility label; the full model adds 0.04.

Two exact equivalences tie these ideas together:

This	is exactly	Consequence
The real-time ±20% bull–bear rule	a hysteresis rule (separate triggers for entering and leaving a state, §5.2): a 20% drawdown from the running peak starts the bear state, a 20% rise from the running trough ends it	its "detections" are threshold crossings, made when the move is already 20% deep
The ex-post ±20% bull–bear label	the real-time rule with each switch moved back to the preceding peak or trough	the two differ only between each turning point and its confirmation: 20% of days since 1927

1.5 Dating, nowcasting and forecasting

Three tasks share the name "regime detection". They use different information and should be evaluated differently.

Task	Information used	Example	Evaluated against	Typical error
Dating	the whole sample, before and after t	NBER recession dates, the ex-post bull–bear label, smoothed HMM probabilities	historical judgement	treating dated labels as observable in real time
Nowcasting	$\mathcal{F}_t$, for a label of day t	"is today in a bear market?", this week's financial-conditions index before it is published	the label, once settled	reporting accuracy on days whose label was already settled
Forecasting	$\mathcal{F}_t$, for a label of a later day	"will next month be high-volatility?"	the realised label	ignoring overlap between consecutive labels

The three tasks correspond to rungs of the filtration ladder in the companion chapter (section 5 of Market Regimes and Hidden Markov Models): dating uses smoothed information, nowcasting filtered information, forecasting predicted information. The most common error in published regime classifiers is to train on dated labels and report accuracy against the same dated labels, as if the accuracy were attainable in real time. It is attainable only on days whose label had already been settled, which are the days where a classifier is least needed.

Nowcasting a label with a long reach is forecasting in disguise. Nowcasting a series that is merely published late is different: the Chicago Fed's National Financial Conditions Index for the week ending Friday is released the following Wednesday, and estimating it on Monday from daily market data is a measurement problem with a short horizon and a known answer. These are good uses of a supervised model, because the target exists and arrives soon.

§1 Key takeaways

1. A regime is a definition chosen to serve a decision. Write the decision, the state variable, the label and the horizon in one sentence before touching data.
2. The reach of a label, the number of future days it depends on, sets what can be learned. A label with zero reach is a rule to compute; a label with positive reach is a forecast.
3. Supervised regime detection is forecasting of the label's future part. Labels whose future part is volatility are learnable; labels whose future part is direction are not.
4. The baseline that matters is the stale label: the most recent fully known label, or the rule's own confirmed state. Judge every classifier by its improvement over it.
5. The ex-post bull–bear label and its real-time version differ only between each turning point and its confirmation. That is a fifth of all days, and it is where the label's value lies.
6. Dating, nowcasting and forecasting are different tasks. Accuracy measured against dated labels on days already settled says nothing about real-time performance.

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2. How practitioners carve up regimes

Practitioners use a dozen regime vocabularies: volatility regimes, bull and bear markets, risk-on and risk-off, stress regimes, business-cycle phases, growth and inflation quadrants, correlation regimes. They look like different ideas. This section argues that they are coarsenings of five state variables, describes each family with the same fields, and shows where the families agree and where they part. The answer to "what kinds of regime can I predict?" falls out of the fields: the families built on the quantity of risk can be forecast, the families built on direction cannot, and the macroeconomic families can be forecast slowly.

2.1 A short lineage

The vocabulary accumulated over a century, and each layer added something a practitioner still uses:

Year	Development	What changed
1922	W. P. Hamilton's *The Stock Market Barometer* sets out Dow theory: primary bull and bear markets, secondary reactions, and confirmation by two averages	The first explicit bull–bear taxonomy, with a confirmation lag built into the definition
1946	Burns and Mitchell, *Measuring Business Cycles* (NBER)	Expansions and contractions dated by expert judgement; the official recession dates that most macro labels still use
1971	Bry and Boschan's turning-point algorithm	Dating becomes a reproducible program; its window and minimum-duration parameters become choices
1989	J. D. Hamilton's Markov-switching model	The regime becomes a latent state with probabilities rather than a dated label
2003–04	Pagan and Sossounov; Lunde and Timmermann	Price-based bull–bear dating with explicit thresholds
2004–12	The Investment Clock; Bridgewater's All Weather	Allocation by growth–inflation quadrant
2006–19	Financial stress and conditions indices from central banks and the OFR	Composite, observable stress regimes published weekly or daily
2018–24	Jump models, clustering, and supervised classifiers trained on model labels	Regimes defined by fitted models; classifiers trained to predict them

The record that matters most for this chapter is the first row. A primary bear market in Dow theory exists only once the averages have confirmed it, so the definition practitioners inherited was a lagging one from the start. The econometric strand of this history, from Hamilton's model onward, is reviewed by Ang and Timmermann (2012).

2.2 Five state variables under every taxonomy

The families below differ in vocabulary, but each one partitions one or two of five continuous state variables. Naming them is the first step to choosing inputs (§3), because inputs are measurements of state variables, not of regimes.

State variable	What it is	Typical observables	Persistence	Known at t?
Quantity of risk	the conditional volatility of returns	realised volatility, VIX, MOVE	high: half-life of one to two months	yes, with small error
Price of risk	compensation demanded per unit of risk; risk appetite	credit spreads, variance risk premium, valuations	high	partly: a spread is observed, its split into expected loss and premium is not
Direction	the conditional mean, or trend	trailing returns, distance from moving averages	low	no: the mean is not identified in short samples
Macroeconomic state	growth, inflation, policy stance	activity data, inflation, the yield curve, the policy rate	very high	with publication lags and revisions
Structure	how assets move together, and how easily they trade	correlations, dispersion, breadth, liquidity measures	moderate	yes, with estimation noise

The column that decides what can be predicted is the last one, together with persistence. The quantity of risk is observable and persistent, so a regime built on it can be forecast a few weeks ahead with real skill. Direction is neither observable nor persistent, so a regime built on it can be dated in hindsight but hardly forecast. The companion chapter shows the statistical root of this asymmetry: in a fitted regime model, state volatilities are pinned down sixteen times more sharply than state means from the same data (section 8 of Market Regimes and Hidden Markov Models). Most practitioner regimes combine

the quantity of risk with direction or with the price of risk. Their observable part is therefore, to a large degree, volatility under another name.

The families that follow are described with the same fields: the **idea**, the **state variables**, the **definitions in use**, typical **durations and episodes**, whether the regime is **observable at** t, the **decision** it serves, its **failure modes**, and **when to use it**.

2.3 Volatility regimes

Idea. Classify the market by how much it moves.

State variables. The quantity of risk.

Definitions in use. Bands on the VIX are the commonest: below 15 "calm", 15–25 "normal", 25–35 "elevated", above 35 or 40 "crisis". These cut points are convention, not estimates **[Practice]**. Since 1990 the VIX's median close has been 17.6; it closed below 15 on 32.5% of days, above 30 on 8% and above 40 on 2.3%. Other definitions use quantiles of realised volatility (the top fifth of the history so far), bands on a GARCH volatility estimate, the calm and turbulent states of a two-state hidden Markov model fitted to returns, and composite scores that average standardised measures of the VIX level, its recent change and its own volatility.

Durations and episodes. Long and numerous. In the worked example the label "next month's realised volatility lands in the top fifth of its history so far" has 166 episodes since 1930, with a median length of 16 days.

Observable at t? The current level, yes. The level over the coming month, which is what a sizing decision needs, is forecastable with an AUC near 0.88 (§6.4).

Decision. Position sizing and volatility targeting, option strategy selection, stop and limit widths, risk budgets.

Failure modes.

- *Non-stationary levels.* A VIX of 20 was a calm reading in 2002, when the index averaged 27, and an alarming one in 2017, when it averaged 11. Fixed bands misclassify whole years.
- *Rolling normalisation turns a level into a surprise.* A score built from a rolling z-score over, say, a year adapts to the level it is measuring. After a year at a VIX of 35 the score reads "normal". Such a score classifies the *change* in risk, which is a different regime from the level of risk, and a sizing rule needs the level. Use long expanding windows or percentiles against the history so far for levels, and reserve short-window z-scores for explicitly measuring surprise (§3.12).
- *Implied is not realised.* The VIX contains a variance risk premium that varies with the price of risk (§3.3), so a VIX band mixes two state variables.
- *A volatility regime is not a direction regime.* High volatility also contains the best days of each decade, which cluster in rebounds. Volatility timing earns its keep by reducing risk, not by avoiding losses. **[Contested]** Whether it also improves returns out of sample is disputed (Moreira & Muir, 2017; Cederburg et al., 2020).

When to use it. Whenever the decision is about risk. It is the default regime, and the only family for which forecasts are reliably better than the stale label.

2.4 Trend regimes: bull, bear, correction, rebound

Idea. Classify the market by the direction and persistence of its price path.

State variables. Direction, with the quantity of risk entering because bear markets are volatile.

Definitions in use. Five constructions are common.

1. *The ±20% rule.* A bear market runs from a peak to the following trough, where the peak is followed by a fall of at least 20% and the trough by a rise of at least 20%. The financial press uses it; Lunde and Timmermann (2004) formalised it with separate thresholds for the two directions. A fall of 10% to 20% is a *correction*.
2. *Business-cycle dating applied to prices.* Pagan and Sossounov (2003) adapted the Bry–Boschan algorithm to monthly stock prices: find local maxima and minima, enforce alternation, censor the ends of the sample, and require a minimum cycle of 16 months and a minimum phase of four months unless the move exceeds 20%.
3. *Trailing-return states.* Cooper, Gutierrez and Hameed (2004) call the market UP when its return over the past 36 months is non-negative. Daniel and Moskowitz (2016) mark a bear state when the past two years' cumulative return is negative.
4. *Moving-average states.* The price is above or below its 200-day or ten-month average (Faber, 2007).
5. *Four-phase cycles.* Goulding, Harvey and Mazzoleni (2023) cross the signs of a slow (12-month) and a fast (1-month) time-series momentum signal: Bull when both are positive, Bear when both are negative, Correction when the slow

signal is positive and the fast one negative, Rebound when the reverse. Maheu, McCurdy and Song (2012) fit a four-state Markov-switching model with bull, bull-correction, bear and bear-rally states.

Durations and episodes. Few. The ±20% rule finds 31 bear markets on the US total market since 1926, nine of them since 2000, with a median length of about six months (128 trading days on the common sample of §6.2; §6.3 lists those since 2000). Longer histories add episodes: Gonzalez, Powell, Shi and Wilson (2005) date two centuries of US bull and bear cycles.

Observable at t? The ex-post definitions (1 and 2) are not: their reach is the confirmation lag, a median of 70 trading days for bear markets. The trailing definitions (3, 4 and 5) are observable, because they are rules with zero reach. They are trend signals rather than forecasts, and the trend-following literature is where their value is assessed (Trend-Following in Financial Markets).

Decision. Market exposure; the risk of momentum strategies, whose crashes cluster in bear states with high volatility, when the market rebounds (Daniel & Moskowitz, 2016); the choice of trend speed (Goulding, Harvey & Mazzoleni, 2023).

Failure modes.

- *Thresholds and index choice flip borderline episodes.* The S&P 500 price index fell 19.9% in 1990, 19.3% in 1998, 19.4% in 2011 and 19.8% in 2018: no bear market by the press convention. The total US market including dividends and small companies fell more than 20% in all four, so each is a bear market in the worked example. Four of its eleven bear markets since 1990 exist or not depending on the index.
- *Bear-market rallies count as bull markets.* The US market rose more than 20% between late November 2008 and early January 2009 before falling to new lows, so the ±20% rule records a brief bull market in the middle of the financial crisis. Four-phase definitions exist to repair this.
- *The confirmation lag.* By construction, the rule recognises a bear market once prices are 20% below the peak.
- *Censoring.* The last phase of any sample is provisional; its end is unknown.

When to use it. For describing history and for benchmarks. As a training target only if the decision really needs direction and the limit on learning direction set out in §1.4 is accepted.

2.5 Trending versus mean-reverting markets

Idea. Classify the market by whether it extends moves or reverses them.

State variables. The persistence of direction: the autocorrelation of returns at the strategy's horizon.

Definitions in use. The variance ratio $\mathrm{VR}(q)$, the variance of q-day returns divided by q times the variance of one-day returns, above one for trending and below one for mean-reverting markets (Lo & MacKinlay, 1988); the Hurst exponent; the autocorrelation of daily returns; and practitioner indicators such as the efficiency ratio (net move divided by path length) and the average directional index **[Practice]**.

Durations and episodes. Short and noisy.

Observable at t? Only with large error. An autocorrelation estimated from 63 days of returns has a standard error of about $1/\sqrt{63} \approx 0.13$, larger than the variation in autocorrelation the regime is meant to detect.

Decision. Choosing between trend-following and mean-reversion strategies; placing stops and targets.

Failure modes. Noise dominates. The P&L of a trend rule depends on the autocorrelation of returns at its own horizon over the holding period, which is a forecast of direction's persistence, not a measurement of it (Trend-Following in Financial Markets).

When to use it. As a descriptive statistic over long windows. There is little out-of-sample support for switching strategies on short-window estimates **[Practice]**.

2.6 Risk-on and risk-off

Idea. On some days and in some periods risky assets rise together while havens fall (risk-on); in others the reverse (risk-off).

State variables. The price of risk, at a global level, and structure.

Definitions in use. The sign or standardised size of a common factor in cross-asset returns, usually the first principal component (the single weighted average of the returns that carries the most common variation) of equities, credit, emerging markets, commodities and carry currencies against Treasuries, gold, the yen, the Swiss franc and the dollar. Chari, Dilts Stedman and Lundblad (2023) build a risk-on/risk-off index from several market-based measures. Miranda-Agrippino and

Rey (2020) find a single global factor behind a large share of the variation in risky-asset prices worldwide. Bekaert, Hoerova and Lo Duca (2013) split the VIX into risk aversion and uncertainty.

Durations and episodes. The daily sign flips often; the strength of the common factor, its share of cross-asset variance over a rolling window, persists for weeks to months.

Observable at t? Yes. It describes today's co-movement.

Decision. Cross-asset hedging, the risk of carry trades and of emerging-market exposure.

Failure modes. The definition is close to circular: a risk-off day is a day on which risky assets fell. The set of havens changes; Treasuries hedged equities from about 2000 to 2021 and did not in 2022 (§2.9). A daily sign is not a regime: use the rolling strength of the factor, or its smoothed level.

When to use it. For multi-asset portfolios, as a structure measure rather than a forecast.

2.7 Financial stress and liquidity

Idea. The financial system either functions normally, or funding and market liquidity are impaired and prices are driven by forced selling.

State variables. The price of risk and structure. In stress, correlations rise and diversification fails.

Definitions in use. Composite indices published by public institutions:

- The **St. Louis Fed Financial Stress Index** (STLFSI4) combines 18 weekly series (seven interest rates, six yield spreads and five other indicators) through their first principal component. Its average since late 1993 is zero, so positive values mean above-average stress (Kliesen & Smith, 2010).
- The **Chicago Fed National Financial Conditions Index** (NFCI) is a weighted average of 105 indicators of risk, credit and leverage, with weights from a dynamic factor model. It is scaled to mean zero and standard deviation one since 1971, and positive values mean tighter-than-average conditions (Brave & Butters, 2012). An adjusted version (ANFCI) removes the part explained by economic activity and inflation.
- The **OFR Financial Stress Index** combines 33 daily market indicators in five categories (credit, equity valuation, funding, safe assets and volatility). It is zero at normal stress, and once estimated its values do not change (Monin, 2019).
- The ECB's **CISS** weights its sub-indices by their cross-correlations, so that stress counts for more when it is systemic, spreading across markets at once (Hollo, Kremer & Lo Duca, 2012).
- **Turbulence**, the squared Mahalanobis distance of today's cross-asset returns from their historical mean (§5.3; Kritzman & Li, 2010), and the **absorption ratio**, the share of variance explained by the leading eigenvectors (Kritzman, Li, Page & Rigobon, 2011), are market-only constructions in the same spirit.

Common cut points are zero (above-average stress), and one and two standard deviations for "significant" and "severe" stress **[Practice]**.

Durations and episodes. Weeks to months, and few systemic episodes: in the US since 1990, roughly 1998, 2001–02, 2007–09, 2011, 2020 and a handful of shorter ones.

Observable at t? Yes for the daily market-based indices, after a lag of a day or two (the OFR index publishes two business days late); with a longer lag for the weekly ones, whose readings for the week ending Friday appear the following Wednesday.

Decision. Leverage, liquidity buffers, risk limits, and whether to trust a diversification assumption.

Failure modes.

- *Revisions.* The NFCI is re-estimated every week, so its history changes. The Chicago Fed describes the changes as usually small, but a backtest on today's history uses weights that were not available then.
- *Full-sample weights.* Weights estimated on the whole sample and applied to its early years carry information from later crises into earlier readings.
- *Composition changes.* The STLFSI was rebuilt as LIBOR was retired: its LIBOR-based spreads gave way to SOFR-based ones, and version 4 replaced a backward-looking 90-day average SOFR with the forward-looking 90-day rate.
- *Conditions are not stress.* The NFCI includes leverage measures, which build slowly in good times; a stress index measures acute strain. The two answer different questions.

Evidence. The NFCI predicts episodes of financial stress at horizons up to a year (Brave & Butters, 2012), and financial conditions forecast the lower tail of GDP growth much better than its median (Adrian, Boyarchenko & Giannone, 2019).

When to use it. For risk limits and leverage, and as a compact input that summarises dozens of series (§3.6).

2.8 Macroeconomic regimes

Idea. Classify periods by the state of the economy, on the theory that asset classes respond to it in understandable ways.

State variables. The macroeconomic state: growth, inflation and policy.

Definitions in use. Four variants.

1. *The business cycle.* NBER expansions and recessions are the reference. Real-time alternatives include Markov-switching dynamic-factor models of activity (Chauvet & Piger, 2008), the Sahm rule, which signals a recession when the three-month average unemployment rate rises half a percentage point above its low of the previous twelve months (Sahm, 2019), and probit models on the yield curve (Estrella & Mishkin, 1998). Hamilton (2011) surveys how such methods have performed in real time, and Giusto and Piger (2017) date turning points in real time with a simple machine-learning classifier, vector quantisation. Composite leading indicators such as the OECD's are usually read in four phases by crossing the indicator's level relative to its trend with its direction: above trend and rising, above and falling, below and falling, below and rising **[Practice]** (OECD, 2012).
2. *Growth–inflation quadrants.* Bridgewater's All Weather framework defines four environments by whether growth and inflation are rising or falling relative to expectations, and holds assets that suit each, balanced by risk (Podolsky, Johnson & Jennings, 2012). Merrill Lynch's Investment Clock names four phases from the direction of growth and inflation: reflation, recovery, overheat and stagflation (Greetham and colleagues, Merrill Lynch, 2004). These are practitioner frameworks from firms that sell allocation products built on them.
3. *Statistical macro-market regimes.* A Gaussian mixture fitted to 17 factor returns from 1971 to 2020 yields four conditions named Crisis, Steady State, Inflation and Walking on Ice (Botte & Bao, 2021). Guidolin and Timmermann (2007) find four states for stocks and bonds jointly: crash, slow growth, bull and recovery.
4. *Monetary-policy regimes.* Hiking, cutting and holding cycles, measured by the six- to twelve-month change in the policy rate or the two-year yield, and the zero-rate period as its own regime.

Durations and episodes. Years, and very few. The US has had 12 recessions since 1948. Inflation regimes number a handful.

Observable at t? Only with publication lags and revisions. NBER dates are not observable in real time at all: the committee announced peaks an average of 7.3 months after they occurred, and troughs 15 months after, over the cycles since 1978 (NBER announcements). The trough of March 1991 was announced in December 1992.

Decision. Strategic allocation, duration and inflation hedging, factor tilts.

Failure modes.

- *Few episodes.* Twelve recessions is the whole post-war sample for a recession classifier.
- *Revisions.* Real-time and final macroeconomic data differ, sometimes enough to change the sign of a growth reading (Croushore & Stark, 2001).
- *Two-sided filters.* A detrended indicator built with the Hodrick–Prescott filter, as the OECD's composite leading indicators have been since 2008, revises its recent values as new data arrive (Hamilton, 2018).
- *Relationships break.* The 10-year minus 3-month Treasury spread was negative from late October 2022 to mid-December 2024, the longest inversion on record, and no NBER recession had followed at the time of writing.

When to use it. For slow allocation decisions, with labels lagged by their publication delay. Rarely for daily equity decisions, where macro data move too slowly to help except through surprises on release days (§3.10).

2.9 Correlation and dispersion regimes

Idea. Classify periods by how assets move together, not by how much they move.

State variables. Structure.

Definitions in use.

- *The stock–bond correlation.* Negative from the late 1990s to about 2020, positive before that and again from 2021–22. Its sign appears to track whether inflation shocks or growth shocks dominate (Campbell, Sunderam & Viceira, 2017); macroeconomic fundamentals explain little of its variation, and liquidity factors more (Baele, Bekaert & Inghelbrecht, 2010). For a balanced portfolio it is the most consequential regime of all (section 14 of Bonds and Bond Markets).
- *Equity correlation.* The average correlation among index members, or the implied correlation from index and single-stock options (Cboe's COR1M and COR3M indices, with history from 2006). **[Contested]** Average correlation predicts market returns, because it tracks aggregate risk (Pollet & Wilson, 2010), and correlation risk carries a premium in index options (Driessen, Maenhout & Vilkov, 2009).

- *Dispersion.* The cross-sectional spread of stock returns separates a "macro" market from a "stock-picker's" market, and is linked to time variation in the value and momentum premiums (Stivers & Sun, 2010).

Durations and episodes. The stock–bond sign changes every decade or two: very few episodes. Equity correlation regimes last weeks to months.

Observable at t? Yes, with estimation noise. A correlation estimated over 63 days has a standard error of roughly $(1 - \rho^2)/\sqrt{63}$, about 0.11 at $\rho = 0.3$. Correlations measured in volatile periods are also biased upward for mechanical reasons (Forbes & Rigobon, 2002).

Decision. Diversification and hedge ratios; dispersion trades; factor exposure.

Failure modes. Estimation noise; the volatility bias; and, for the stock–bond sign, a sample of two or three regimes in a century.

When to use it. Whenever a portfolio relies on a correlation. It is the regime most often ignored and most often decisive.

2.10 Calendar and event regimes

Some regimes are known in advance: Federal Reserve announcement days, the release days of inflation and employment data, monthly options expiry, quarter ends, earnings seasons, holidays. Volatility and volume behave differently on them. They should enter a study as features, never as something to detect. Leaving them out has a cost: a classifier that does not know a release is scheduled may read the release-day spike in volatility as the start of a regime.

2.11 How the families overlap, and where they part

The families overlap because volatility rises when prices fall. Low volatility, bull markets, risk-on, low stress and expansions tend to occur together. The overlap is real but partial. In the worked example's century of US data, the chance-corrected agreement between the ex-post bear-market label and the next-month high-volatility label is $\kappa = 0.35$, between either and NBER recessions about 0.3, and between the rule's own real-time bear state and a negative next quarter 0.03 (§6.2).

The episodes where the families part are the informative ones. The table uses the worked example's data: the US total market, the VIX and the Baa–Treasury credit spread.

Episode	Largest drawdown in the window	VIX	Baa – 10-year spread, start → high	NBER recession	Families that fired
Aug–Dec 1987	33.1%	not yet published	2.0 → 2.7 points	no	volatility, trend; no macro
Jan–Jun 2007	5.9%	mean 13, high 20	flat near 1.6	no	almost none; the yield curve was inverted
Jul–Dec 2007	9.7%	mean 22, high 31	1.6 → 2.6 points	no (the peak was December)	credit and stress before equity
Sep–Dec 2018	20.4%	high 36	1.9 → 2.5 points	no	trend, volatility; mild credit
Feb–Apr 2020	34.2%	high 83	2.1 → 4.3 points	yes	all of them, briefly
Jan–Oct 2022	25.4%	mean 26, high 36.5	1.8 → 2.4 points	no	trend; stock–bond correlation turned positive
Feb–Apr 2025	19.6%	high 52	1.5 → 2.0 points	no	volatility; trend on some indices

Two lessons follow. First, a single "regime" label hides which family fired, and the families call for different responses: the credit-led stress of late 2007 called for cutting leverage while equity volatility was still moderate, and the 2022 bear market called for hedges other than Treasuries. Second, the borderline cases decide the labels: 2025 is a bear market on some indices (the Nasdaq Composite fell more than 20%) and not on the total market, which fell 19.6%, and the decision a study takes on such cases shapes everything trained on them.

2.12 The taxonomy as a product space

The families are coordinates in the master form of §1.3, not branches of a tree:

Family	State variable	Usual K	Usual dating	Reach of the usual label	US episodes	Decision
Volatility	quantity of risk	2–4	threshold or quantile	0 (level) or 21 days (forward)	over a hundred	sizing, risk
Bull–bear	direction	2–4	turning points	median 70 days for bear onsets	31 since 1926	exposure
Trending vs mean-reverting	direction's persistence	2	rolling statistic	0 (estimated)	noisy	strategy choice
Risk-on/off	price of risk, structure	2	factor sign or strength	0	many, short	cross-asset hedging
Stress	price of risk, structure	2–3	composite index	0, plus publication lag	about ten	leverage, limits
Business cycle	macro	2–4	expert or model	months to years	12 since 1948	allocation
Growth–inflation	macro	4	sign of changes	months	a handful	allocation
Correlation	structure	2	rolling estimate	0 (estimated)	2–3 for stock–bond	diversification

Which axis matters. The state variable chosen determines what can be forecast far more than the details of the definition. Within a family, thresholds change the episode count a great deal and the labels much less: moving from ±20% to ±15% raises the number of bear markets from 31 to 47, yet the two labels agree with $\kappa = 0.88$. Across families, agreement falls to between 0.03 and 0.58. Spend design effort on choosing the state variable and the horizon; tune thresholds last.

§2 Key takeaways

1. Practitioner regimes are coarsenings of five state variables: the quantity of risk, the price of risk, direction, the macroeconomic state and structure. Name the state variable before choosing inputs.
2. Regimes built on the quantity of risk are observable and persistent, so they can be forecast. Regimes built on direction can be dated in hindsight and hardly forecast. Macro regimes move slowly and arrive late.
3. The ±20% bull–bear convention is index-dependent at the margin: four of the eleven bear markets on the total US market since 1990 were not bear markets on the S&P 500 price index.
4. Stress and conditions indices are good compact inputs, but the weekly ones are revised and use full-sample weights. Prefer indices that are not revised, or store the vintages.
5. Macro regimes have a dozen episodes and labels published months late. Lag them by their real announcement delay or do not use them.
6. The families overlap because volatility rises when prices fall, and they part in exactly the episodes that matter. Keep them as separate outputs rather than merging them into one label.
7. Thresholds move episode counts more than they move labels. Choose the state variable and horizon first; tune thresholds last.

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3. Inputs: observing the state variables

An input is useful to a regime classifier only through what it says about a state variable. This section goes through the input families by the state variable they measure, gives each one's mechanism, history and failure modes, and ends with the two practical problems that decide whether any of it works: aligning inputs to the time they became known, and transforming them so that a relationship estimated in one decade holds in the next. The section's conclusion, borne out by the worked example, is that the input space is much smaller than the list of series suggests: two factors carry most of the variation, and the price path itself carries most of the information.

3.1 What makes a good regime input

Six properties, in rough order of importance:

1. **Mechanism.** A reason the input moves with the state variable the label depends on. A credit spread widens with the price of default risk; the VIX rises with expected volatility. An input with no mechanism is a specification search waiting to happen.
2. **History.** Enough history to contain many episodes of the regime. An input that starts in 2009 has never seen 2008, and in the worked example a model that gained a new input but lost nineteen years of history did worse (§6.4).
3. **Timeliness.** Published soon after the fact, at a frequency at least as high as the decision's. A monthly series cannot inform a daily decision except on release days.
4. **Point-in-time integrity.** Never revised, or revised with stored vintages. A revised series used at its latest vintage leaks future information into the past.
5. **Stationarity, or a transform that delivers it.** A level that drifts across decades must be converted into something comparable over time (§3.12).
6. **Non-redundancy.** Information beyond the inputs already present. Most candidate inputs are noisy measurements of the volatility factor (§3.13).

Recommendation. Rank candidate inputs by these properties before testing any of them. Testing forty inputs against a label with thirty episodes finds something every time, and the something does not survive.

3.2 The price path itself

What it measures. The quantity of risk and the direction, directly.

The inputs. Realised volatility over several horizons (5, 21 and 63 days), the downside semivariance, the drawdown from the running or one-year high, trailing returns over one, three and twelve months, the distance from moving averages, realised skewness, and jump measures. With intraday data, realised variance from five-minute returns is a far more precise volatility measurement than a daily-return estimate (Andersen, Bollerslev, Diebold & Labys, 2003), and the heterogeneous autoregression of daily, weekly and monthly realised variance is the standard forecasting model built on it (Corsi, 2009).

Why they come first. They are available for the longest history, they are never revised, and they measure the two state variables most labels depend on. In the worked example, seven price-path features alone reached an AUC of 0.87 for next month's volatility and 0.83 for the ex-post bear-market label, and no other input set beat them by more than 0.01 (§6.4).

Failure modes. Price-path features of direction (trailing returns, distance from averages) are rules with zero reach. They describe the trend that has happened; any forecasting value they have is the time-series momentum effect, with its own literature and its own crashes (Moskowitz, Ooi & Pedersen, 2012).

3.3 The implied-volatility complex

What it measures. The quantity of risk expected by option buyers, plus the premium they pay for it.

The inputs.

- **The VIX**, the 30-day model-free implied volatility of the S&P 500, with history from 1990 (Cboe VIX methodology). Its square is the risk-neutral expectation of the next month's variance, which equals the physical expectation plus the variance risk premium (sections 5 and 6 of Implied Volatility).
- **The VIX term structure.** Cboe publishes 9-day (VIX9D, from 2011), 3-month (VIX3M; its free history file starts in September 2009) and 6-month (VIX6M, from 2008) indices. The ratio of the VIX to the VIX3M is normally below one; above one, the curve is inverted, which marks acute stress. It was inverted on 7.7% of days from September 2009 to 2025, in 99 separate spells. The slope carries information about the returns of volatility assets, such as variance swaps and VIX futures (Johnson, 2017; Simon & Campasano, 2014).
- **The variance risk premium**: the VIX squared minus a forecast of realised variance, often proxied by trailing realised variance. **[Contested]** It predicts equity returns, most strongly at about a quarter's horizon (Bollerslev, Tauchen & Zhou, 2009). Splitting the VIX into the premium and the expected variance, the premium predicts returns and the expected variance predicts economic activity and financial instability (Bekaert & Hoerova, 2014) **[Contested]** on magnitude and stability.
- **Volatility of volatility and tails.** The VVIX (the VIX of the VIX, from 2006) and the SKEW index (the price of S&P 500 tail protection, from 1990). Evidence that the SKEW anticipates crashes is weak; it measures demand for protection, which is high in calm, complacent markets as well as before falls **[Contested]**.

- **Cross-asset implied volatility.** The MOVE index of Treasury implied volatility, crude oil (OVX), gold (GVZ), the Nasdaq-100 (VXN) and the Russell 2000 (RVX), and implied volatility on credit ETFs. These measure the quantity of risk in other markets and help when the label is about those markets.

Why it matters. The VIX is forward-looking where realised volatility is backward-looking. In the worked example the VIX alone forecast next month's volatility better than trailing realised volatility alone, with an AUC of 0.87 against 0.84 (§6.4).

Failure modes.

- *Non-stationary levels* (§2.3).
- *A premium that moves.* The VIX mixes expected volatility with the price of volatility risk; when the premium jumps, a VIX band misreads expected risk.
- *Flicker in term-structure ratios.* Ninety-nine inversion spells in sixteen years is a signal that switches on and off within days. Use a smoothed ratio or hysteresis.
- *Short histories.* Every term-structure input except the VIX itself starts after 2005 in free data, so a model that relies on them is trained on two or three stress episodes.

3.4 Credit

What it measures. The price of default risk, and through it the price of risk in general.

The inputs.

- **Moody's Baa yield minus the 10-year Treasury yield**, daily on FRED from 1986 (series BAA10Y). Long-history monthly versions extend the comparison back much further.
- **Option-adjusted spreads of the ICE BofA high-yield and investment-grade indices.** Since April 2026 FRED publishes only a rolling three-year window of these, so a long history has to be stored as it is fetched or bought.
- **The Gilchrist–Zakrajšek credit spread and the excess bond premium**, the part of corporate spreads not explained by expected default. The premium predicts economic activity better than the spread itself (Gilchrist & Zakrajšek, 2012); the Federal Reserve publishes it monthly from 1973 (Favara, Gilchrist, Lewis & Zakrajšek, 2016).
- **Credit ETF prices and options**: high-yield and investment-grade bond funds, the ratio of a high-yield fund to a Treasury fund, and implied volatility on the high-yield fund.

Mechanism. A spread is expected default loss plus a risk premium plus a liquidity premium (sections 8 and 9 of Bonds and Bond Markets). The premium component moves with the price of risk, which is why credit can lead equity: in the second half of 2007 the Baa spread widened by a full percentage point while the stock market's largest fall was under 10% (§2.11).

Failure modes. Composition drift: index ratings mixes and durations change over decades. Level shifts across eras: a Baa spread of 2.5 points was a stress reading in 2018 and an ordinary one in parts of the 1980s. Duration mismatch: the Baa yield refers to long bonds, compared here with a 10-year Treasury.

3.5 Rates and the yield curve

What it measures. The macroeconomic state and expected policy.

The inputs. The 10-year minus 3-month Treasury spread (daily from 1982), the 10-year minus 2-year spread, the near-term forward spread, real yields and breakeven inflation rates from inflation-protected Treasuries, the 2-year yield as a proxy for expected policy, and the MOVE index.

Evidence. The term spread is the best single predictor of recessions two to six quarters ahead (Estrella & Hardouvelis, 1991; Estrella & Mishkin, 1998) **[Fact]** for the post-war sample. A systematic comparison of leading indicators finds the term spread strongest four to six quarters ahead, with its recent path and broker-dealer margin balances adding information at other horizons (Liu & Moench, 2016). The spread between the expected short rate six quarters ahead and the current short rate is a cleaner version of the signal (Engstrom & Sharpe, 2019).

For equity regimes. The curve is slow. In the worked example the 10-year minus 3-month spread was nearly uncorrelated with the volatility inputs (correlation 0.05 with the log VIX) and added nothing to daily classifiers (§3.13, §6.4).

Failure modes. The relationship rests on eight or so recessions since the 1960s, and its latest signal has not been followed by one so far: the inversion from October 2022 to December 2024 was the longest on record (§2.8). The slope mixes expected policy with a term premium that quantitative easing compressed, and the zero lower bound pinned the short end from 2009 to 2015.

3.6 Composite stress and conditions indices

What they measure. The price of risk and structure, summarised from dozens of series.

The inputs. The STLFSI4 and NFCI (weekly), the OFR Financial Stress Index (daily), and the ECB's CISS for the euro area (§2.7). Each is a factor model that someone else maintains: it compresses many spreads, rates and volatilities into one number.

Use them with three precautions. Lag each index to its publication date: weekly readings for the week ending Friday are released the following Wednesday. Prefer indices that are not revised, such as the OFR index, or store the vintages of those that are, such as the NFCI (FRED's archival service, ALFRED, keeps them). And remember that weights estimated on the full sample make the index's early history partly hindsight.

3.7 Breadth and market internals

What they measure. Structure: how widely a move is shared.

The inputs. The share of index members above their 50- and 200-day moving averages; the advance–decline line; new 52-week highs minus new lows; up and down volume; the NYSE TICK (stocks trading on an uptick minus those on a downtick, intraday); the Arms index or TRIN; and ratios of equal-weight to cap-weight indices, small to large companies, and cyclical to defensive sectors.

Mechanism. Narrowing participation is held to precede market tops, and broad "breadth thrusts" to follow bottoms **[Practice]**.

Evidence. Breadth and similar sentiment measures move strongly with contemporaneous returns but have little power to predict near-term returns (Brown & Cliff, 2004).

Failure modes.

- *Survivorship.* Breadth computed from today's index members excludes the stocks that were removed after falling. Historical breadth then looks healthier than it was in every downturn, and a classifier trained on it learns a false relationship. Use point-in-time membership, or treat the bias as known and avoid breadth levels.
- *Data quality.* Exchange-published breadth series have gaps and definitional changes; derived series need consistent universes.
- *Noise.* TICK and TRIN are intraday microstructure measures. At daily horizons they are dominated by noise.

3.8 Correlation, dispersion and cross-asset structure

What they measure. Structure, and through it the price of risk.

The inputs. The average pairwise correlation of index members, or each member's correlation with the index, over one to three months; implied correlation (COR1M and COR3M); the cross-sectional dispersion of returns; the rolling stock–bond correlation; turbulence and the absorption ratio (§2.7); and the strength of a cross-asset risk factor (§2.6).

Evidence. Average correlation predicts market returns (Pollet & Wilson, 2010), and correlation risk is priced in index options (Driessen, Maenhout & Vilkov, 2009). Turbulence and the absorption ratio rise before and during systemic episodes (Kritzman & Li, 2010; Kritzman, Li, Page & Rigobon, 2011) **[Contested]**: the authors' firm sells analytics built on them, and the evidence is largely in-sample.

Failure modes. Estimation noise (§2.9). The volatility bias: correlation rises mechanically with volatility, so a correlation regime can be a volatility regime in disguise (Forbes & Rigobon, 2002). Survivorship in member-based measures, as with breadth.

3.9 Sentiment and positioning

What they measure. The price of risk as revealed by what investors pay for protection or how they are positioned.

The inputs. Put/call volume ratios: the *equity* ratio, dominated by trading in single-stock options; the *index* ratio, dominated by institutional hedging and structurally near or above one; and the total. Survey sentiment (the American Association of Individual Investors' weekly survey), futures positioning reported by the CFTC, margin debt, fund flows, and composite indices such as that of Baker and Wurgler (2006).

Evidence. Extreme readings of the VIX, the put/call ratio and TRIN have been found to have contrarian predictive power for S&P 500 futures returns (Simon & Wiggins, 2001). Other work finds that such sentiment measures are caused by past returns and volatility and add little once those are controlled for (Wang, Keswani & Taylor, 2006). Where option volume does

predict returns, the information is in the volume of buyers opening new positions, which is not public; the public ratios do not carry it (Pan & Poteshman, 2006). The evidence for public sentiment measures as regime inputs is **[Contested]** at best.

Structural breaks. The composition of option volume has changed enough to break the comparability of put/call ratios over time. Retail option trading grew sharply from 2020 (Bryzgalova, Pavlova & Sikorskaya, 2023), and options expiring the same day grew from a small fraction of S&P 500 index option volume a decade ago to more than 60% in May 2025 (Cboe, 2025). A level of the index put/call ratio in 2025 does not mean what the same level meant in 2015.

Recommendation. If you use sentiment inputs, use them as residuals: regress each on recent returns and volatility, and keep what those do not explain. Then test the residual against the price-path baseline. Expect little.

3.10 Macroeconomic data

What it measures. The macroeconomic state.

The inputs. Weekly initial unemployment claims; monthly payrolls, purchasing managers' indices, industrial production and retail sales; composite activity indices such as the Chicago Fed National Activity Index, and the daily business-conditions index of Aruoba, Diebold and Scotti (2009); inflation measures; and **surprise indices**, which standardise each release's deviation from the consensus forecast and aggregate across releases (Scotti, 2016). Any economic calendar that records consensus forecasts alongside actual values supports a surprise index.

Real time. Macro data are revised, sometimes heavily, and the vintage available on a date differs from today's history (Croushore & Stark, 2001). Daily nowcasting models update an estimate of the current quarter as each release arrives (Giannone, Reichlin & Small, 2008).

For equity regimes. Macro data move the slow regimes of §2.8. For daily equity classification they matter mainly through surprises on release days.

3.11 Information timing

A daily regime study joins series that are observed at different times, published with different lags, and revised on different schedules. The join is where most regime studies leak. The rule is simple to state: **align every input to the time it became available, not to the date it describes.**

Input	Describes	Usable from	Revised?
Index closes, VIX family	the trading day	that day's close	no
Treasury yields and the Baa spread (FRED)	a business day	the next business day	no
ICE BofA option-adjusted spreads (FRED)	a business day	the next morning	no
NFCI, STLFSI4	the week ending Friday	the following Wednesday	NFCI weekly; check vintages
OFR Financial Stress Index	a business day	two business days later	no
Cboe put/call ratios	the trading day	that evening, after the close	no
Initial claims	the week ending Saturday	the following Thursday	yes
Payrolls, CPI	the month	the following month	payrolls twice, then annually; CPI rarely
NBER turning points	a month	4 to 21 months later	rarely
Breadth from index members	the trading day	that day	survivorship unless point-in-time

Four rules follow:

1. **Lag to publication.** Shift each series by its release lag before joining it to the price data.
2. **Store what you fetched, when you fetched it.** A store that overwrites history with the latest vintage destroys the evidence of what was known.
3. **Forward-fill weekly and monthly series from their release date**, never from their reference date.
4. **Never normalise with full-sample statistics.** A z-score that uses the full-sample mean and standard deviation carries the future into every past value.

3.12 Transforming inputs

Raw levels rarely survive a change of decade. A model trained when the VIX averaged 27 and applied when it averages 11 extrapolates. Transforms make inputs comparable across time:

- **Logs** for volatilities and ratios of volatilities. The log VIX is closer to symmetric and mean-reverting than the VIX itself.
- **Ratios and differences** for term structures (VIX/VIX3M) and for the gap between implied and realised volatility.
- **Changes** over one to three months for slow-moving levels such as spreads and the curve.
- **Percentiles or z-scores against an expanding window** for levels whose meaning should be absolute: a VIX in the top decile of all history is high whatever the last year was like.
- **Z-scores against a rolling window** only for inputs whose meaning is relative to recent experience. A rolling z-score measures surprise. Applied to a persistent level it erases the level, and a regime score built that way classifies changes in risk rather than risk (§2.3).
- **Clipping** at four or five standard deviations, so that one crisis does not dominate the fit and a new crisis does not produce an absurd extrapolation.

Every transform must be causal: computed from data up to t only. An expanding quantile threshold satisfies this; a full-sample quantile does not.

3.13 How many independent inputs are there?

Fewer than the list suggests. On daily data from September 2009 to 2025, seven standardised inputs (the log VIX, log 21-day realised volatility, the log VIX/VIX3M ratio, the Baa spread, the 10-year minus 3-month spread, the drawdown from the one-year high, and the distance from the 200-day average) have a first principal component that explains 54% of their joint variance and a second that explains 22%. On the six with history back to 1990, the shares are 62% and 19%. The correlations show the structure:

	log VIX	log RV	VIX/VIX3M	Baa spread	10y–3m	drawdown	trend
log VIX	1.00						
log realised vol	0.81	1.00					
VIX/VIX3M	0.68	0.55	1.00				
Baa spread	0.32	0.29	0.12	1.00			
10y–3m spread	0.05	−0.03	0.01	0.58	1.00		
drawdown	0.74	0.75	0.57	0.22	−0.12	1.00	
trend	−0.52	−0.60	−0.48	−0.28	0.05	−0.85	1.00

Two clusters stand out: a fast **volatility and drawdown** factor (VIX, realised volatility, term structure, drawdown, trend) and a slow **credit and curve** factor. Adding a twentieth series usually adds a noisier measurement of the first factor, not a new dimension.

Recommendation. Build one or two composite scores per state variable, each from a few well-understood inputs, and give those to the classifier. The classifier then has five to ten inputs, which its episode count can support.

3.14 A data map

The table lists free sources and the depth of their daily history, which sets how many episodes a classifier can learn from.

Input	State variable	Free source	Daily history from	Published	Revised
US market total return	quantity of risk, direction	Kenneth French data library	1926	monthly batches	rarely
VIX	quantity of risk	Cboe	1990	same day	no
VIX9D, VIX3M, VIX6M	term structure	Cboe	2011, 2009, 2008	same day	no
VVIX, SKEW	volatility of volatility, tails	Cboe	2006, 1990	same day	no
COR1M, COR3M	implied correlation	Cboe	2006	same day	no

Input	State variable	Free source	Daily history from	Published	Revised
MOVE	rates volatility	ICE, via data vendors	1988	same day	no
Baa minus 10-year	credit	FRED	1986	next day	no
High-yield and investment-grade OAS	credit	FRED (three-year window)	rolling	next day	no
Excess bond premium	price of risk	Federal Reserve	1973 (monthly)	monthly	re-estimated
10y–3m and 10y–2y spreads	macro, policy	FRED	1982, 1976	next day	no
NFCI, ANFCI	conditions	Chicago Fed, FRED	1971 (weekly)	Wednesday	weekly
STLFSI4	stress	FRED	1993 (weekly)	Wednesday	check vintages
OFR Financial Stress Index	stress	OFR	2000	two days later	no
Put/call ratios	sentiment	Cboe	2006	same evening	no
NBER recession months	macro label	NBER, FRED	1854 (monthly)	months later	rarely

Two gaps recur. Long, free histories of high-yield spreads no longer exist on FRED, so a study that needs them must store its own copy or use the Baa spread and the excess bond premium. And the volatility term structure, implied correlation and most cross-asset volatility indices start between 2006 and 2011, so a model that uses them has seen only two or three stress episodes. Several of those indices were also launched years after the start of their published history, and the earlier values are back-calculations from archived option prices: point-in-time in their inputs, but never published at the time.

§3 Key takeaways

1. Choose inputs by the state variable the label depends on. For risk labels that means volatility inputs first; for macro labels, the curve and credit.
2. The price path is the first input, not the last. It has the longest history, is never revised, and in the worked example no other input set beat it by more than 0.01 of AUC.
3. The VIX is the best single forward-looking volatility input. The term structure, implied correlation and cross-asset volatility indices add little on average and have short histories.
4. Credit can lead equity in credit-led stress, as in 2007. It is the main reason to include a second factor.
5. Public sentiment measures, including put/call ratios, are mostly a reflection of past returns, and their levels have structural breaks. Residualise them or leave them out.
6. Align every input to its publication time, store vintages, forward-fill from release dates, and never normalise with full-sample statistics.
7. Rolling z-scores measure surprise, not level. Use expanding percentiles for inputs whose level matters.
8. The input space has about two dimensions: a fast volatility factor and a slow credit-and-curve factor. Five to ten well-chosen inputs are enough.

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4. Labels: where the training targets come from

A supervised regime classifier is only as good as the label it is trained to predict, and the label is the part of a regime study that receives the least thought. This section treats labels as objects with measurable properties. It starts from one decomposition, surveys six sources of labels with a fixed set of fields, and ends with a checklist for vetting a label and a procedure for building training and test sets that never use a label before it was knowable.

4.1 The label master form

Every label is a rule applied to a window of data around its date:

$$y_t = L(\underbrace{S_{t-a}, \dots, S_t}_{\text{past part: in } \mathcal{F}_t},\ \underbrace{S_{t+1}, \dots, S_{t+b}}_{\text{future part}}).$$

The past part is observable when the decision is made; the future part is not. Three properties of every label follow from its reach b, and they are the first things to compute about a new label:

1. **The purge.** At a refit date τ, the labels of the last b days before τ are not yet known. Training on them uses information from after τ.
2. **The detection floor.** A classifier can know the future part only by forecasting it. For labels whose future part is direction, that forecast is weak, so detection lags by roughly the reach.
3. **Censoring.** The last b days of any sample have no label, or a provisional one. In a live system this is the present, where the labels are always missing.

A label is a proxy for an outcome that a decision cares about. It is a good label if it separates that outcome, if it persists over the decision's horizon, if it does not change much when its parameters change slightly, if it has enough episodes to learn from, and if its reach leaves something to learn. §4.9 turns these into tests.

The six sources below are described with the same fields: the **idea**, the **definition**, the **reach**, the **parameters**, the **episodes** and **persistence** it typically produces, its **stability**, its **decision relevance**, its **failure modes**, and **when to use it**.

4.2 Contemporaneous rules

Idea. Label a day by a condition on data observed that day.

Definition. "The VIX closed above 30"; "the market closed at least 10% below its high"; "the NFCI is above zero"; "the price is below its 200-day average"; "trailing 21-day realised volatility is in the top fifth of its history".

Reach. Zero. The label is in $\mathcal{F}_t$.

Parameters. A threshold, sometimes a window.

Episodes and persistence. Many, and often too many: a contemporaneous rule flickers when its input hovers near the threshold. In the worked example the rule "10% below the high" produced 132 episodes since 1930 with a median length of three days (§6.2). Hysteresis, with separate thresholds for entering and leaving, removes most of the flicker (§5.2).

Stability. Usually moderate; the threshold matters.

Decision relevance. Direct when the decision itself is defined by the condition, as with a risk limit that binds when the VIX exceeds 30.

Failure modes. Training a model to predict a label with zero reach learns nothing that the rule does not already compute. Two supervised uses remain legitimate. The first is *nowcasting a late-published rule*: estimate this week's NFCI from daily market data before Wednesday, or the monthly Sahm indicator from weekly claims. The second is *forecasting the rule h days ahead*, which turns it into a forward-outcome label (§4.5).

When to use it. As a regime when the decision needs only the current state; as a feature; and as a baseline for everything else.

4.3 Official and expert dating

Idea. Use the turning points published by an authority.

Definition. NBER business-cycle peaks and troughs; the OECD's turning points in its composite leading indicators; crisis chronologies; the bear-market dates reported by the financial press.

Reach. Long and variable: the announcement lag. The NBER announced peaks an average of 7.3 months after they occurred and troughs 15 months after, over the cycles since 1978. The longest wait was 21 months, for the trough of March 1991.

Parameters. None that the user controls; the committee's judgement is the parameter.

Episodes and persistence. Very few and very persistent: 12 US recessions since 1948, each lasting months.

Stability. High: dates are rarely revised.

Decision relevance. High when a mandate or a contract keys off the official event; otherwise indirect.

Failure modes. The dates are set with hindsight. A classifier trained on them learns to recognise a whole recession, including its first months, which at the time looked like an ordinary slowdown. Used without the announcement lag, the label is a look-ahead, and a backtest that conditions on it reports a fantasy (section 1.5 of Market Regimes and Hidden Markov Models).

When to use it. As a benchmark, and as a target only for slow macro decisions, lagged by the real announcement delay.

4.4 Turning-point rules on the price path

Idea. Date peaks and troughs from the price path itself, by rule.

Definition. The $\pm\delta$ rule of §2.4: a peak is confirmed when the price falls δ below its running high, a trough when it rises δ above its running low, and the label marks the days between a peak and the following trough. Variants include the Bry–Boschan and Pagan–Sossounov algorithms (§2.4), within the general framework for dating cycles from turning points of Harding and Pagan (2002), drawdown-based phases, *directional-change* event series, which record each reversal of a fixed size and so measure time in price moves rather than days (Glattfelder, Dupuis & Olsen, 2011; Tsang, Tao, Serguieva & Ma, 2017), and the zigzag indicator of charting software **[Practice]**.

Reach. Random: the confirmation lag. For the ±20% rule on the US market since 1926 the median lag was 70 trading days for bear markets (interquartile range 42–146) and 41 for bull markets (16–76). For ±15% the medians were 48 and 25 days.

An exact equivalence makes the structure plain. The rule's real-time version is a hysteresis rule on drawdown and run-up. The ex-post label is that rule's state with each switch moved back to the extreme that preceded it. The two labels differ only on the days between each turning point and its confirmation, which in the worked example are 20% of all days since 1927.

Two-sided smoothers used to define trends or cycles, such as centred moving averages, the Hodrick–Prescott filter and band-pass filters, have the same look-ahead in continuous form. The smoothed value at t depends on data after t, and the values near the end of the sample are revised as data arrive (Hamilton, 2018).

Parameters. The threshold δ, or the window and minimum durations of a dating algorithm.

Episodes and persistence. Few and persistent: 31 bear markets at ±20% since 1926, 47 at ±15%, nine at ±20% since 2000.

Stability. High agreement, unstable counts. The ±20% and ±15% labels agree with $\kappa = 0.88$, but the second has half again as many episodes and shorter confirmation lags.

Decision relevance. High for exposure decisions in principle. In practice the valuable part of the label is the unknowable part (§6.3).

Failure modes. Borderline episodes flip with the index and the threshold (§2.4). Bear-market rallies become bull markets. The final phase is censored: on any date, the labels after the most recent running extreme are unknown, and training on them leaks. The label rewards forecasting turning points, which is forecasting direction.

When to use it. For describing history and for benchmarking. As a training target only with the purge of §4.10 applied, and with the expectation that the classifier will mostly learn to read the drawdown. The literature comparing the two ways of defining bull and bear markets reaches a consistent split: rule-based dating describes them better in sample, and Markov-switching models forecast them better (Kole & van Dijk, 2017); using several markets' returns jointly helps both (Liu, Maheu & Song, 2024).

4.5 Forward-outcome labels

Idea. Define the regime by what happens over a fixed window after t, chosen to match the decision.

Definition. Several forms are common.

- *Forward volatility quantile.* $y_t = 1$ if realised volatility over $(t, t+h]$ exceeds q_t, where q_t is a quantile of trailing h-day realised volatility over the history up to t. With $h = 21$ and the 80th percentile this is the "next month's volatility in the top fifth" label of the worked example.
- *Forward drawdown event.* $y_t = 1$ if the largest fall within $(t, t+h]$, measured from S_t or any later peak in the window, exceeds a level such as 10%.
- *Forward return sign or quantile.* $y_t = 1$ if $S_{t+h} < S_t$.
- *Triple barrier.* Place an upper and a lower barrier, usually scaled by current volatility, and a time limit; the label records which is hit first (López de Prado, 2018).
- *Trend scanning.* Fit linear trends over forward windows of several lengths and label by the sign of the slope with the largest t-statistic (López de Prado, 2020).

Reach. Fixed at h, or bounded by the time limit.

Parameters. The horizon and the threshold. The threshold must be computed causally, with an expanding quantile rather than a quantile of the full sample, because a full-sample quantile carries information about the future into every label.

Episodes and persistence. Depends on what the future part measures. In the worked example the forward-volatility label had 166 episodes since 1930 (median length 16 days), the forward-drawdown label 71 (median 60 days), and the forward-return label 501, with a median length of four days: a direction label flickers because direction does not persist.

Stability. Good for volatility labels; poor for return labels.

Decision relevance. The highest of any source, because the label is the decision's outcome by construction.

Failure modes.

- *Overlap.* Consecutive labels share $h-1$ days of outcome, so neighbouring training rows are nearly copies of each other. Standard cross-validation, which scatters neighbours across folds, then measures memorisation. Purging (dropping the training rows whose label windows overlap the test block) and an embargo of at least h days (dropping the rows just after it) fix this (López de Prado, 2018; section 11 of Market Regimes and Hidden Markov Models).
- *Full-sample thresholds*, as above.
- *Horizon mismatch*: a label with a one-month horizon trained for a decision held for a week.

When to use it. Whenever the decision can be written as a forward event. It is the default recommendation of this chapter.

4.6 Labels from a fitted model

Idea. Fit an unsupervised regime model to the history, and use its state sequence as the label.

Definition. The smoothed state probabilities of a hidden Markov model, computed by the backward smoother of Kim (1994), used as soft labels; the most likely state path; the state sequence of a **statistical jump model**, which minimises the distance of each day's features from its state's centroid plus a penalty λ for every switch (Bemporad, Breschi, Piga & Boyd, 2018; Nystrup, Lindström & Madsen, 2020; Aydınhan, Kolm, Mulvey & Shu, 2024); clusters from k-means or a Gaussian mixture (Botte & Bao, 2021), including clustering of whole return distributions with the Wasserstein distance (Horvath, Issa & Muguruza, 2024); and the segments of a change-point method.

Reach. Formally the whole sample, since a two-sided fit lets every observation influence every label. In practice the label of day t is settled by the data within a few persistence lengths of t. The worked example measures it for a two-state jump model on exponentially weighted features with half-lives of 10 and 42 days and $\lambda = 40$. The model's forward-only state path, which uses no data after t, agreed with its two-sided labels on 94% of days ($\kappa = 0.84$), and recognised entry into the high-risk state a median of eight trading days late (interquartile range 6–11). That is a reach of about a week and a half, against months for the ±20% rule.

A second leak sits in the parameters. A model fitted to the whole sample defines its states using data from after t, even when its state assignment is causal. Refit the labelling model on data up to each refit date.

Parameters. The number of states, the features, and the persistence penalty or transition prior. The penalty sets the episode count directly: λ of 10, 40 and 100 gave 112, 54 and 29 high-risk episodes on the century of data, with median lengths of 30, 68 and 132 trading days.

Episodes and persistence. Chosen through the penalty.

Stability. Moderate. Labels near the end of a fitting window are revised as data arrive, and states can swap identities between refits; identify each state by a fixed property, such as having the lower mean return or the higher volatility.

Decision relevance. Indirect. The model decides what a regime is, and a likelihood or a clustering objective usually decides it is volatility. That suits risk decisions and not much else.

Failure modes. The two leaks above; a definition chosen by the fitting objective rather than by the decision; and the false comfort of persistent labels, which make a classifier's accuracy look high because yesterday's label predicts today's.

When to use it. When no decision-defined label exists and a persistent summary of many features is wanted, and as the teacher in a teacher–student pipeline (§5.8).

4.7 Strategy-outcome labels

Idea. Label a day by which of several actions would have done best after it.

Definition. Which of a set of strategies earned the highest risk-adjusted return over $(t, t+h]$; whether a base strategy's trade opened at t made money, the **meta-label** of López de Prado (2018); whether a risky asset beat a haven over the next month.

Reach. h.

Parameters. The set of strategies, the horizon, and the scoring rule.

Episodes and persistence. As noisy as the return differences they compare.

Stability. Low when strategies' returns are close.

Decision relevance. Maximal for strategy selection: the label is the decision.

Failure modes. A strategy-outcome label is a direction label in disguise, because it depends on future returns. Choosing among many strategies inflates the apparent skill of the selector, and the comparison must be made net of costs.

When to use it. When the regime study exists to switch between strategies. The meta-label is the better-behaved form: it asks only whether to act on a signal you already have, which is a narrower question than which strategy is best.

4.8 Comparison of label sources

The fields that discriminate, in one table. The learnability ratings are this chapter's assessment, informed by the worked example.

Source	Reach	What a classifier can learn	Typical US episodes	Persistence	Main failure
Contemporaneous rule	0	nothing: compute it	many	low without hysteresis	flicker
Official dating	months to years	slow macro forecasts	12 recessions since 1948	very high	used without the announcement lag
Turning-point rule	random, median 70 days at bear onsets	mostly the current drawdown	31 bears since 1926	high	the value lies in the unknowable days
Forward outcome, volatility	h	a great deal (AUC near 0.88)	over a hundred	moderate	overlap ignored in validation
Forward outcome, drawdown	h	some (AUC near 0.65)	dozens	moderate	overlap; few severe events
Forward outcome, return	h	almost nothing (AUC near 0.45)	hundreds of flickers	very low	mistaken for a regime
Fitted model	whole sample, about eight days in practice	the model's volatility state	set by the penalty	high	two-sided labels, full-sample parameters
Strategy outcome	h	as much as the return gap allows	depends	low	a direction label in disguise

4.9 Vetting a label before training anything

Five checks, each with a number to compute. A label that fails one of them will not be rescued by a better model.

1. **Decision relevance.** Compare the outcome the decision cares about across the label's classes: the strategy's forward Sharpe ratio, the forward volatility, the forward drawdown. If the classes do not separate it, the label is the wrong label. A volatility label that separates forward volatility but not the strategy's returns serves a sizing decision and not a timing one.
2. **Episodes and persistence.** Count the episodes and their median length. The median length should exceed the decision's holding period, and the episodes should number at least a few dozen in the training set. Fewer than about twenty training episodes leaves too little to learn from **[Practice]**; the test period needs its own episodes as well (§4.10).
3. **Stability.** Recompute the label with its parameters moved by a fifth to a half (the threshold from 20% to 15%, the horizon from 21 to 15 or 30 days) and measure Cohen's kappa against the original. Kappa below about 0.7 means the label is mostly its parameters **[Practice]**.
4. **Reach and censoring.** Compute the reach: fixed, or the distribution of confirmation lags. It sets the purge, the embargo and the expected detection lag.
5. **The value of knowing it, and when.** Compute what the decision would earn with the label known one day ahead (the oracle) and with the label known when it becomes known (the stale label). The oracle is the ceiling; the stale label is the floor any classifier must clear; the gap is what a classifier is for. For the ±20% bull–bear label the ceiling was a Sharpe ratio of 1.54 and the floor 0.46, against 0.45 for doing nothing (§6.3). A classifier that recovers a fifth of that gap would be remarkable; one that recovers none is the usual result.

4.10 Building the training and test sets

The procedure below prevents the leaks specific to regime labels. Each step names the leak it closes.

1. **Point-in-time features.** Align every input to its publication time and use stored vintages for revised series (§3.11). *Closes: look-ahead in the inputs.*
2. **Causal label parameters.** Compute thresholds with expanding windows. The outcome of a forward label lies in the future by design; its threshold must not. *Closes: look-ahead in the definition.*
3. **Train only on labels final at the refit date.** At refit date τ, include day t only if its label was settled by τ: $t \leq \tau - b$ for a fixed reach; t up to the current running extreme (the latest high in a bull phase, the latest low in a bear phase) for a

turning-point label; for a model-derived label, refit the labelling model on data up to τ and use only labels older than its effective reach. *Closes: the purge.*

4. **Walk forward.** Refit on an expanding or rolling window at a fixed cadence (quarterly or yearly) and predict the next period. When data are too scarce for that, use purged k-fold cross-validation with an embargo of at least the reach. *Closes: evaluation on the training period.*
5. **Keep episodes whole.** Never let one episode contribute rows to both the training and the test side of a split. With overlapping labels, a split in the middle of an episode puts nearly identical rows on both sides. *Closes: memorisation scored as skill.*
6. **Weight for overlap.** Down-weight rows whose labels share outcomes, using the average uniqueness of each label (López de Prado, 2018), or keep every b-th row. This does not change the point estimates of a well-specified model much, but it stops the effective sample size from being overstated by a factor of b.
7. **Leave class balance alone if you need probabilities.** Oversampling the rare class distorts predicted probabilities. Use class weights only if the decision uses ranks rather than probabilities, and recalibrate afterwards.
8. **Size the test set by episodes.** A test period should contain at least ten episodes of the minority class, and results should be reported per episode as well as per day.
9. **Prefer soft labels where they exist.** Smoothed state probabilities carry the labelling model's own uncertainty, and training against them, as in knowledge distillation (Hinton, Vinyals & Dean, 2015), wastes less information than training against their rounded values.
10. **Know where the label noise is.** Classifiers tolerate label noise that is unrelated to the features (Frénay & Verleysen, 2014). Regime label noise is not of that kind: it concentrates at transitions, exactly where the features are changing. Expect a classifier to be least reliable at the moments that matter most.

§4 Key takeaways

1. Every label is a rule on a window of data around its date. Compute its reach first: it sets the purge, the detection floor and the censoring.
2. Contemporaneous rules have zero reach. Compute them; train a model on them only to nowcast a late-published value or to forecast the rule ahead.
3. Official and expert dates are hindsight. Lag them by their real announcement delay or do not train on them.
4. Turning-point labels are a real-time hysteresis rule with its switches back-dated. The back-dated days are a fifth of history and hold all of the label's value.
5. Forward-outcome labels are the default: they are the decision's outcome by construction. Their thresholds must be causal and their overlap purged.
6. Model-derived labels inherit two leaks: two-sided state assignment and full-sample parameters. Refit the labeller at every refit date.
7. Vet a label before modelling it: decision relevance, episodes, stability, reach, and the gap between the oracle and the stale label.
8. Train only on labels that were final at the refit date, keep episodes whole across splits, and size test sets by episodes.

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5. Classifiers

With inputs chosen and a label vetted, the remaining choice is the model. It matters less than the two choices before it. In the worked example, going from a one-input to a twelve-input model added 0.04 of AUC on the volatility label (0.84 to 0.88), while moving from the volatility label to a direction label cost 0.44 (0.88 to 0.44), eleven times as much (§6.4). This section surveys the models with a fixed set of fields: the **idea**, the **definition**, the **assumptions** (and which are false in practice), **strengths**, **weaknesses**, **cost**, **robustness**, **failure modes** and **when to reach for it**. It begins with the baselines, because a classifier is only as good as its margin over them, and ends with the post-processing that turns a flickering probability into a usable regime.

5.1 The baselines every classifier must beat

Two baselines matter, and a classifier should be scored against both.

Climatology predicts the base rate of the label in the training data, the same probability every day. Its skill is zero by definition, and the Brier skill score (§7.1) measures improvement over it: $\text{BSS} = 1 - \text{BS}_{\text{model}} / \text{BS}_{\text{climatology}}$.

The stale label predicts the current label from the most recent label already known (§1.4): trailing realised volatility for a forward-volatility label, the confirmed state for a turning-point label. It captures persistence, and persistence is most of what any regime classifier exploits. In the worked example the stale label for next month's volatility already had a Brier skill score of 0.34 and an AUC of 0.84 (§6.4).

Recommendation. Report skill over climatology for comparability with the literature, and skill over the stale label for the decision. A model that does not beat the stale label is a slow way to compute it.

5.2 Threshold rules with hysteresis

Idea. Switch regime when an observable crosses a threshold, with a gap between the entry and exit thresholds so the regime does not flicker.

Definition. For a single input x_t and thresholds $u_{\text{out}} < u_{\text{in}}$:

$$s_t = \begin{cases} 1 & \text{if } x_t \geq u_{\text{in}}, \\ 0 & \text{if } x_t < u_{\text{out}}, \\ s_{t-1} & \text{otherwise.} \end{cases}$$

The regime enters at u_{in} and stays until the input falls below u_{out}; with $u_{\text{out}} = u_{\text{in}}$ the rule is a plain threshold and flickers. The real-time ±20% bull–bear rule is this rule with the watched input changing with the state: the drawdown from the running peak starts the bear state at 20%, and the run-up from the running trough ends it at 20%.

Assumptions. The regime is a monotone function of one observable. True for volatility regimes; false for most others.

Strengths. Transparent, with no estimation and no training labels; easy to monitor; hard to overfit.

Weaknesses. One input; fixed thresholds that age badly when the input's level drifts (§2.3).

Cost. Negligible.

Robustness. High to noise, given hysteresis; low to level drift.

Failure modes. Thresholds set on the full sample leak; thresholds that are never revisited misclassify whole decades; without hysteresis, flicker.

When to reach for it. When the regime *is* the observable, as with "VIX above 30", and as the baseline for every other model.

5.3 Composite indices

Idea. Combine several standardised inputs into one score, then threshold the score.

Definition. With $\tilde{x}_{j,t}$ the causally standardised j-th input, signed so that a higher value means more of the state variable (more risk, more stress),

$$I_t = \sum_{j=1}^{d} w_j \, \tilde{x}_{j,t},$$

with weights w_j that are equal, taken from the first principal component, from a dynamic factor model (the NFCI), or from cross-correlations (the CISS). Turbulence is a quadratic cousin that uses the inverse covariance matrix: with R_t the vector of returns on m assets, and their mean $\bar{R}$ and covariance Σ estimated from data up to t, $D_t = (R_t - \bar{R})^\top \Sigma^{-1} (R_t - \bar{R})$. This is the squared Mahalanobis distance of today's returns from their mean: for one asset, the squared z-score; in general, large on days when assets move unusually far *or* in unusual combinations.

Assumptions. The inputs share one dominant factor and the factor is the state variable of interest. For volatility and stress this holds well (§3.13).

Strengths. Uses many inputs with few parameters; averages away input-specific noise; needs no labels.

Weaknesses. It summarises co-movement among the inputs, not the outcome a decision cares about, so its relationship to the label must still be checked.

Cost. Low.

Robustness. Equal weights are more robust than estimated ones when episodes are few **[Practice]**.

Failure modes. Weights or normalisations estimated on the full sample; an input whose behaviour changes (the put/call ratio, §3.9) silently changing the index; rolling standardisation that turns the index into a surprise measure.

When to reach for it. To build the state scores of §3.13 that a classifier then uses, or as the regime itself when no label is available.

5.4 Logistic and probit models

Idea. Model the probability of the label as a smooth, monotone function of a weighted sum of inputs.

Definition. For a binary label,

$$\Pr(y_t = 1 \mid x_t) = \Lambda(\beta_0 + \beta^\top x_t), \qquad \Lambda(u) = \frac{1}{1 + e^{-u}},$$

fitted by maximum likelihood with a ridge penalty: on standardised inputs, the coefficients minimise the negative log-likelihood summed over the n training rows plus $\frac{1}{2}\,\alpha\, n \|\beta\|^2$, with the intercept β_0 left unpenalised. Dividing by n, this is the mean negative log-likelihood plus $\frac{1}{2}\,\alpha\|\beta\|^2$, so α means the same at any sample size. The penalty shrinks the coefficients towards zero: at $\alpha = 0$ the fit is plain maximum likelihood, and as α grows the model collapses to climatology ($\beta = 0$, the base rate).

Three variants. The probit model replaces Λ with the normal distribution function and behaves almost identically. For ordered regimes (low, normal, high, crisis), the ordered logit uses one coefficient vector and $K - 1$ increasing cut points γ_k: $\Pr(s_t \le k \mid x_t) = \Lambda(\gamma_k - \beta^\top x_t)$. **Dynamic** binary models add the lagged label or a lagged latent index to the right-hand side, and are the standard tools for recession and bear-market forecasting (Kauppi & Saikkonen, 2008; Nyberg, 2013; Chen, 2009).

Assumptions. The log-odds of the label are linear in the inputs. The assumption is false at the extremes, which is why inputs are logged, clipped and standardised first.

Strengths. Calibrated probabilities when the model is roughly right; few parameters, so it is stable with few episodes; coefficients a human can read.

Weaknesses. No interactions unless added by hand; monotone in each input.

Cost. Negligible.

Robustness. Good with shrinkage, poor without it. In the worked example, with a penalty of $\alpha = 0.001$ per training row, the twelve-input model scored an AUC of 0.51 on the drawdown label, against 0.66 for a one-input model fitted the same way; with $\alpha = 0.1$ it scored 0.64. Shrinkage matters because the effective sample is the number of episodes, not of rows.

Failure modes. In a dynamic model, a lagged label used without lagging it by its reach and publication delay: a lagged NBER indicator is not known at $t - 1$. Extrapolation: a linear model meeting an input outside its training range produces extreme probabilities.

When to reach for it. By default. **Recommendation:** start every supervised regime study with a regularised logistic model on five to ten transformed inputs, and require any more complex model to beat it out of sample.

5.5 Tree ensembles

Idea. Combine many decision trees, each splitting the input space at thresholds, to capture non-linear effects and interactions.

Definition. Random forests average deep trees fitted to bootstrap samples; gradient boosting fits shallow trees in sequence, each to the errors of the ensemble so far.

Assumptions. Enough examples near each split to estimate it. With regime labels, "examples" means episodes.

Strengths. Interactions such as "high volatility *and* a downtrend" without specifying them; insensitivity to monotone transforms of the inputs; tolerance of uninformative inputs (Grinsztajn, Oyallon & Varoquaux, 2022).

Weaknesses. More parameters to fit from the same few episodes; predicted probabilities that need recalibration; and no extrapolation. A tree treats a VIX of 80 like the highest VIX in its training data.

Cost. Moderate; trivial at daily frequency.

Robustness. Sensitive to the depth, the learning rate and the number of trees; tune them with purged validation.

Failure modes. Overfitting to a handful of crises that dominate the high-volatility region of the input space; a stepwise response that flips the regime at an arbitrary split.

When to reach for it. For labels with a hundred or more episodes, such as volatility labels, and for heterogeneous inputs. It is the classifier of the best-known teacher–student study (§5.8).

5.6 Sequence models

Idea. Let a recurrent or attention-based network learn its own features from the raw history of the inputs.

Definition. Recurrent networks, temporal convolutional networks and transformers mapping a window of past inputs to label probabilities.

Assumptions. Enough data to learn features that hand-made ones miss.

Strengths. No feature engineering; can find temporal patterns.

Weaknesses. Data-hungry, with a few dozen episodes to learn from; opaque; unstable across training runs.

Cost. High.

Robustness. Low with small samples.

Failure modes. Memorising the few crises in the training data; leakage through normalisation layers fitted on the full sample.

When to reach for it. Rarely, for regime labels. The most convincing use of learned macroeconomic states is inside a larger asset-pricing objective rather than as a standalone regime classifier (Chen, Pelger & Zhu, 2024). A network that does not beat a regularised logistic model with good inputs is the expected outcome **[Practice]**.

5.7 Latent-state models with observable drivers

Idea. Skip the labels: let a hidden Markov model or Markov-switching model infer the regime from the inputs, with persistence built in.

Definition. A hidden Markov model whose emissions are the multivariate inputs; a Markov-switching model whose transition probabilities depend on observables (Diebold, Lee & Weinbach, 1994; Filardo, 1994); a Markov-switching dynamic factor model of economic activity (Chauvet & Piger, 2008). The companion chapter covers estimation in detail (section 5 of Market Regimes and Hidden Markov Models).

Assumptions. A small number of states with constant emission distributions and Markov persistence.

Strengths. Probabilities and persistence come with the model; no labels to construct; a predictive density.

Weaknesses. The fitted states mean whatever the likelihood prefers, usually volatility; transition probabilities are estimated from the few transitions in the sample.

Cost. Moderate.

Robustness. Improved by heavy-tailed emissions and persistence priors (section 6 of Market Regimes and Hidden Markov Models).

Failure modes. Smoothed probabilities used as real-time signals; states that change meaning between refits.

When to reach for it. When no external label suits the decision and a probabilistic, persistent state is wanted. Its filter is also the post-processor of §5.11, where it smooths another model's output.

5.8 Teacher–student pipelines

Idea. Use an unsupervised model to define the regimes (the teacher), then train a supervised classifier on causal inputs to predict them (the student).

Definition. Fit a statistical jump model or hidden Markov model to features of past returns. Its two-sided state sequence becomes the label. Train a classifier, often gradient boosting, on inputs known at t, including cross-asset and macroeconomic data the teacher never saw. Shu, Yu and Mulvey (2024b) label bullish and bearish regimes for each of twelve assets with jump models, forecast them with gradient-boosted trees using asset-specific and cross-asset macro features, and report better portfolios across several allocation methods from 1991 to 2023 **[Contested]**: one study, by the method's developers, and at odds with the small gains from cross-asset inputs in the worked example (§6.4). The same group uses jump-model

regimes for downside-risk reduction (Shu, Yu & Mulvey, 2024a) and for factor allocation (Shu & Mulvey, 2024). Earlier studies used the same design with other teachers. Hauptmann, Hoppenkamps, Min, Ramsauer and Zagst (2014) forecast the filtered probabilities of a two-regime Markov-switching model of monthly S&P 500 returns with logistic regressions. Uysal and Mulvey (2021) estimate the probability of a recession or a stock-market contraction with random forests on a large macroeconomic database, out of sample from 1973.

Assumptions. The teacher's regimes matter for the decision, and the student's inputs carry information about the teacher's labels beyond what the teacher's own causal state already has.

Strengths. Persistent labels with no external definition; access to the full supervised toolbox; the student can use inputs the teacher could not model.

Weaknesses. Two leaks inherited from model-derived labels (§4.6): two-sided state assignment and full-sample parameters. The student learns to predict the teacher, not the outcome; if the teacher's states are volatility states, the student is a volatility forecaster with extra steps.

Cost. Moderate.

Robustness. Depends on the teacher's stability across refits.

Failure modes. A student scored against labels from a teacher fitted on the whole sample, which overstates its accuracy near transitions. Labels near each refit date that the next refit revises.

When to reach for it. When a persistent regime definition is wanted and labels must come from somewhere. The procedure: refit the teacher at each refit date on data up to that date; train the student only on labels older than the teacher's effective reach; score the student against the teacher's own causal state as the stale-label baseline. The method is a form of knowledge distillation (Hinton, Vinyals & Dean, 2015), and training against the teacher's soft probabilities rather than its hard states keeps the teacher's uncertainty.

5.9 Change-point alarms

Idea. Detect that the data-generating process has changed, without saying which of K known regimes it changed to.

Definition. Cumulative-sum (CUSUM) detectors accumulate evidence of a shift in mean or variance and alarm at a threshold. Bayesian online change-point detection maintains a posterior over the time since the last change, the run length, updated with each observation (Adams & MacKay, 2007).

Assumptions. Segments are internally stable and changes are abrupt.

Strengths. Answers "did something just change?", which is often the operative question; explicit control of false alarms; no labels.

Weaknesses. Says nothing about what the new state is; non-recurrent by design.

Cost. Low to moderate.

Robustness. Sensitive to the assumed size of the change and to the prior on run length.

Failure modes. Alarms on every volatility burst in fat-tailed data unless the observation model has heavy tails.

When to reach for it. As a monitoring layer beside a classifier: an alarm that the inputs' behaviour has shifted, prompting review or a reduction in size.

5.10 Skipping the regime

Idea. Forecast the quantity the decision needs directly, and discretise only for communication, if at all.

Definition. For a sizing decision, forecast next month's volatility with a regression on trailing realised volatility at daily, weekly and monthly scales (Corsi, 2009) and on the VIX, and size inversely to the forecast. For a drawdown decision, forecast a low quantile of next quarter's return with quantile regression on stress inputs, as growth-at-risk does for GDP (Adrian, Boyarchenko & Giannone, 2019).

Assumptions. The decision is a smooth function of the forecast.

Strengths. No information is thrown away by coarsening; no label construction; the forecast can be evaluated directly.

Weaknesses. Less interpretable than a named regime; discrete decisions still need a threshold somewhere.

Cost. Low.

Robustness. High.

Failure modes. The same input-timing leaks as everything else.

When to reach for it. First. **Recommendation:** build the continuous forecast before the regime classifier, and adopt a discrete regime only where it improves the decision over the forecast it coarsens. The companion chapter reaches the same conclusion from the latent-state side: size continuously, do not gate (section 12 of Market Regimes and Hidden Markov Models).

5.11 Post-processing: calibration, smoothing and hysteresis

A classifier's daily probability is noisy, and a regime should not change every time it crosses one half. Four post-processing steps, in increasing order of structure:

1. **Calibration.** Map raw scores to probabilities that match observed frequencies, by fitting a logistic curve to the scores (Platt scaling) or a monotone step function (isotonic regression). Fit it on training-period data only, and for a flexible model on out-of-fold scores, because such a model's scores on its own training rows are too confident. Tree ensembles need it most (Niculescu-Mizil & Caruana, 2005).
2. **Smoothing.** An exponentially weighted average of the probability trades a few days of delay for stability.
3. **Hysteresis.** Enter the regime when the probability exceeds, say, 0.6, and leave when it falls below 0.4 (§5.2).
4. **A Markov filter over the classifier.** Treat the classifier's output as evidence and run it through a persistent Markov chain. With π_k the share of class k in the classifier's training data and p_{jk} the probability of moving from class j to class k in a day, the filtered belief $\tilde{\xi}_t$ updates as

$$\tilde{\xi}_t^{(k)} \;\propto\; \Big(\sum_j p_{jk}\, \tilde{\xi}_{t-1}^{(j)} \Big) \frac{\hat{p}_t^{(k)}}{\pi_k}.$$

   The ratio $\hat{p}_t^{(k)}/\pi_k$ is the classifier's evidence for class k, with its training prior divided out; the sum carries yesterday's belief forward through the chain. This is the Hamilton filter (section 5 of Market Regimes and Hidden Markov Models) with the classifier in the place of the emission density, and it is how speech recognition combined neural networks with hidden Markov models (Bourlard & Morgan, 1994). The stickiness of p_{jk} sets the trade between delay and flicker: with no persistence ($p_{jk} = \pi_k$ for every j) the filter returns the classifier's own probability, and the stickier the chain, the more consecutive evidence it takes to move the belief. The filter treats each day's evidence as independent given the state, which features built from overlapping windows are not, so read its output as a smoothed score rather than a calibrated probability.

For more than two ordered classes, smooth the cumulative probabilities rather than each class separately, so that the classes remain ordered.

5.12 Comparison

The ratings are this chapter's assessment.

Method	Needs labels	Output	Persistence	Episodes needed	Main failure	Reach for it when
Threshold with hysteresis	no	state	via hysteresis	none	level drift	the regime is the observable
Composite index	no	score	via smoothing	none	full-sample weights	building state scores
Logistic or probit	yes	probability	via inputs or a filter	dozens	extrapolation; unpurged lags	the default supervised model
Tree ensemble	yes	score	via a filter	a hundred or more	few episodes per split	many episodes, many inputs
Sequence network	yes	probability	learned	many hundreds	memorising crises	seldom

Method	Needs labels	Output	Persistence	Episodes needed	Main failure	Reach for it when
Latent-state model	no	probability	Markov	dozens of transitions	states mean volatility	no external label fits
Teacher–student	from a model	probability	from the teacher	as the student needs	two leaks from the teacher	a persistent definition is wanted
Change-point alarm	no	run length	by construction	none	fat-tail false alarms	monitoring
Direct forecast	no	quantity	via inputs	none	input leaks	first, always

§5 Key takeaways

1. Judge every classifier against the stale label as well as against the base rate. Persistence is most of what any regime classifier exploits.
2. Start with a regularised logistic model on five to ten transformed inputs. Shrinkage matters because the effective sample is the number of episodes.
3. Tree ensembles earn their complexity only on labels with a hundred or more episodes. Sequence networks rarely do.
4. Composite indices are the right way to use many inputs: as a few state scores feeding a small classifier.
5. Teacher–student pipelines inherit their teacher's leaks. Refit the teacher at each refit date and score the student against the teacher's causal state.
6. Forecast the decision variable directly before building a regime, and keep the regime only if it improves the decision.
7. Post-process probabilities: calibrate, then smooth with hysteresis or a Markov filter whose stickiness sets the trade between delay and flicker.

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6. A worked example on a century of data

This section runs the chapter's argument on public data. It builds eight definitions of a "bad" regime on a century of US stock returns, measures how much they agree, prices the hindsight in the most popular one, and then trains causal classifiers with and without cross-asset inputs to see what each kind of label allows them to learn. Every number is printed by `figures/rc_common.py`; the figures are drawn by `figures/rc_labels.py` and `figures/rc_classifier.py`.

6.1 Data and protocol

Data. The price path is the daily total return of the value-weighted US stock market from the Kenneth French data library, July 1926 to December 2025, 26,151 trading days. Until 1952 the library includes Saturday sessions, so the early years have up to about 300 trading days a year, and a 252-day "year" is shorter there. The VIX (from 1990) and the VIX3M (from September 2009) come from Cboe's public history files. The Baa–Treasury spread (BAA10Y, from 1986), the 10-year minus 3-month Treasury spread (T10Y3M, from 1982) and the NBER recession indicator (USREC) come from FRED. The two FRED spreads are lagged one day, to their publication.

Labels. Nine labels, each marking a "bad" class: eight definitions and the ±15% variant of the first.

1. *Bear, ±20% rule, ex post*: days from a peak to the following trough, where peaks and troughs are confirmed by 20% moves (§4.4).
2. *Bear, ±20% rule, as confirmed*: the same rule's real-time state.
3. *Bear, ±15% rule, ex post*: as the first, with 15% thresholds.
4. *10% below the high, now*: the market closed at least 10% below its all-time high.
5. *Next-month volatility, top fifth*: realised volatility over the next 21 trading days at or above the 80th percentile of trailing 21-day realised volatility up to t, an expanding threshold used after a burn-in of 1,260 trading days (five years at 252 a year).
6. *10% drawdown in next quarter*: a fall of at least 10% within the next 63 trading days.
7. *Negative next-quarter return*: the market lower 63 trading days later.

8. *Jump model, high-risk state*: the two-sided state sequence of a two-state statistical jump model with $\lambda = 40$, fitted on the data from 1927 on. Its features are exponentially weighted means and downside deviations of daily returns, with half-lives of 10 and 42 days.
9. *NBER recession*: months the NBER dates as recession, without any announcement lag; it is used here only to compare with the others.

Classifiers. Logistic regressions (§5.4) with a ridge penalty of $\alpha = 0.1$ per training row on standardised inputs, refitted every January from 2000 to 2025. Each refit uses all rows since January 1990 whose label was final by the refit date: rows older than the label's reach for forward labels, and rows up to the last confirmed extreme for the ex-post bear label. Four input sets are compared. *Vol-only* is the log of trailing 21-day realised volatility, the stale label for the volatility target. *VIX-only* is the log VIX. *Price* has seven price-path features: log realised volatility over 5, 21 and 63 days, the drawdown from the one-year high, the 12-month and 1-month log returns, and the log distance from the 200-day average. *Price + cross-asset*, the full model, adds five more: the log VIX, a variance-risk-premium proxy (the log VIX minus log realised volatility), the Baa spread, its 21-day change, and the 10-year minus 3-month spread.

Scores. AUC and the Brier skill score (§5.1, §7.1, §7.3) against climatology, the base rate in each refit's training data, on the 2000–2025 test predictions. An AUC of 0.5 is chance, and a Brier skill of 0 is no better than climatology. Differences between models come with 90% intervals from a bootstrap that resamples one-year blocks of the test period.

6.2 Eight definitions, one century: how much they agree

The labels are compared on the days where all of them are defined, October 1930 to October 2025 (24,809 days). The start is set by the 1,260-day burn-in of the volatility threshold, the end by the 63-day forward windows. The ±20% rule finds 31 bear markets since 1926; the first ends before the common sample begins, so the table counts 30. For the same reason it counts 45 of the 47 bear markets of the ±15% rule.

Label	Share of days in the bad class	Episodes	Median length (days)	Reach
Bear, ±20% rule, ex post	19%	30	128	confirmation lag, median 70 at bear onsets
Bear, ±20% rule, as confirmed	14%	30	93	0
Bear, ±15% rule, ex post	20%	45	71	confirmation lag, median 48 at bear onsets
10% below the high, now	40%	132	3	0
Next-month volatility, top fifth	17%	166	16	21 days
10% drawdown in next quarter	21%	71	60	63 days
Negative next-quarter return	32%	501	4	63 days
Jump model, high-risk state	24%	52	68	whole sample; about eight days in practice
NBER recession	16%	15	209	months to years

The next table gives Cohen's kappa for each pair of labels, on the same days. Kappa corrects the raw share of agreement for the agreement that two labels with these base rates would reach by chance (Cohen, 1960):

$$\kappa = \frac{\text{agreement observed} - \text{agreement by chance}}{1 - \text{agreement by chance}}.$$

Two labels with base rates π_A and π_B agree by chance on a share $\pi_A\pi_B + (1-\pi_A)(1-\pi_B)$ of days. A kappa of 1 is perfect agreement and 0 is no better than chance.

	1	2	3	4	5	6	7	8	9
1. Bear, ±20%, ex post	1.00								
2. Bear, ±20%, as confirmed	0.27	1.00							
3. Bear, ±15%, ex post	0.88	0.18	1.00						
4. 10% below the high	0.33	0.38	0.32	1.00					
5. Next-month volatility	0.35	0.24	0.33	0.21	1.00				
6. Drawdown next quarter	0.58	0.21	0.54	0.23	0.38	1.00			

	1	2	3	4	5	6	7	8	9
7. Negative next quarter	0.38	0.03	0.38	0.08	0.11	0.52	1.00		
8. Jump model	0.50	0.47	0.45	0.47	0.50	0.37	0.11	1.00	
9. NBER recession	0.30	0.30	0.29	0.25	0.27	0.24	0.08	0.33	1.00

Four readings:

1. **Definitions in different families agree only moderately.** Apart from the ±20% and ±15% versions of the ex-post bear label, no pair reaches 0.6. There is no single regime history that all reasonable definitions recover.
2. **Thresholds matter less than hindsight.** The ±20% and ±15% ex-post labels agree at 0.88. The ±20% ex-post label and its own real-time version agree at 0.27.
3. **The ex-post bear label is mostly about the future.** It agrees better with *future* outcomes than with any rule that uses only the past: 0.58 with a 10% drawdown in the next quarter, 0.38 with a negative next quarter and 0.35 with high volatility next month, against 0.27 with its own real-time version and 0.33 with 10% below the high. The real-time version agrees with a negative next quarter at 0.03, which is to say not at all.
4. **The jump model sits in the middle.** It agrees with almost everything at about 0.5 and with the direction label at 0.11, as a volatility state should.

Figure 1 shows eight of them, all but the ±15% variant, from 1995 to 2025.

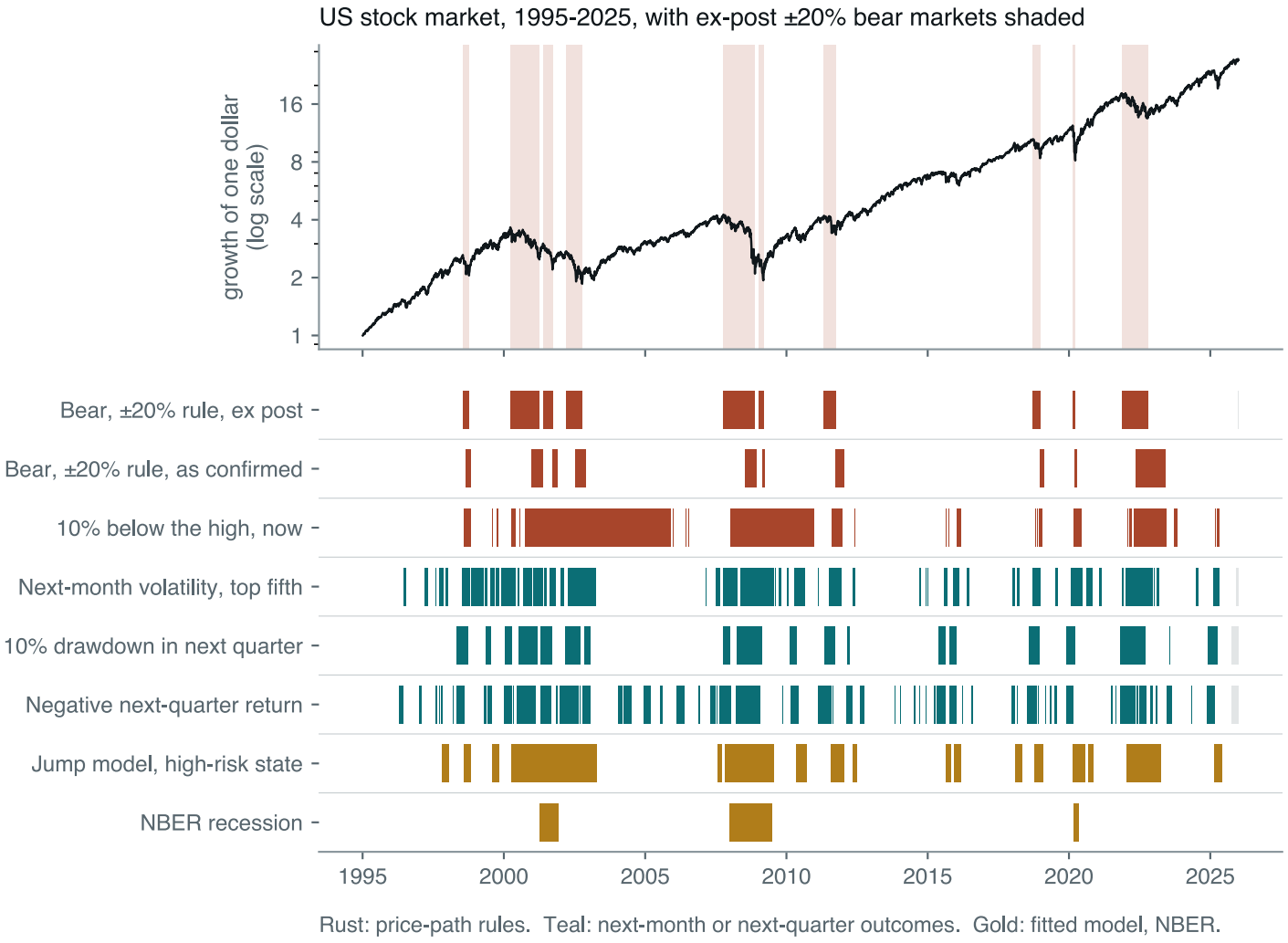


Figure 1. Eight definitions of a bad regime, 1995–2025. Rust strips are rules on the price path, teal strips outcomes over the next month or quarter, gold strips a fitted model and the NBER. Grey at the right edge marks the final weeks, where forward labels are not yet defined.

6.3 The price of hindsight: the ±20% rule

The ±20% rule is the most widely quoted regime definition, and its real-time version can be compared with its ex-post version exactly.

Confirmation lags. Since 1926, a bear market was confirmed a median of 70 trading days after its peak (interquartile range 42–146), and a bull market a median of 41 trading days after its trough (16–76). With 15% thresholds the medians fall to 48 and 25. On 20% of all days since 1927, the confirmed state and the ex-post label disagree: on 12.7% of days the label is bear while the confirmed state still says bull, and on 7.4% the reverse.

The bear markets since 2000 show the range:

Peak	Confirmed	Trading days to confirmation	Fall at confirmation
24 Mar 2000	20 Dec 2000	188	21.9%
21 May 2001	17 Sep 2001	78	21.1%
19 Mar 2002	10 Jul 2002	78	20.2%
9 Oct 2007	15 Jul 2008	192	20.3%
6 Jan 2009	27 Feb 2009	36	20.4%
29 Apr 2011	3 Oct 2011	108	20.3%
20 Sep 2018	24 Dec 2018	65	20.4%
19 Feb 2020	12 Mar 2020	16	27.5%
8 Nov 2021	19 May 2022	133	20.0%

In 2011 and 2018 the bear market was confirmed on the day of its trough: the rule recognised the bear market on the day it ended. The 2020 bear market fell 27.5% in the sixteen days the rule needed.

What the label is worth, and when. Three ways to hold the US market or cash (the risk-free rate) from 1927 to 2025. The Sharpe ratio is computed from daily returns in excess of the risk-free rate, and the maximum drawdown is the largest peak-to-trough fall in wealth:

Strategy	Compound annual return	Sharpe ratio	Maximum drawdown
Buy and hold	9.7%	0.45	84.1%
Out of the market in ex-post bear markets (the oracle)	24.2%	1.54	19.6%
Out of the market when the rule has confirmed a bear market	8.7%	0.46	61.3%

The oracle needs to know on each day whether that day belongs to a bear market, which is known only after confirmation. Using the rule's state from the previous day instead, as a real system must, leaves the Sharpe ratio where buy and hold had it. The real-time rule does reduce the worst drawdown, from 84% to 61%, because it exits the deepest bear markets after their first 20%. The difference between 1.54 and 0.46 is the value of the unknowable 20% of days. It is the entire prize that a bull–bear classifier competes for, and it can be captured only by forecasting falls of 20% before they have happened.

6.4 What a causal classifier learns

The classifiers are trained on four of the labels: next-month volatility, a 10% drawdown in the next quarter, a negative next-quarter return and the ex-post ±20% bear label. Their future parts differ: volatility for the first, volatility and direction for the second, direction for the third, and for the fourth the current drawdown plus the direction of the coming months. Their out-of-sample scores, 2000–2025, as AUC with the Brier skill score against climatology in brackets:

Target	Base rate	Vol-only	VIX-only	Price	Price + cross-asset
Next-month volatility, top fifth	0.32	0.84 (+0.34)	0.87 (+0.38)	0.87 (+0.45)	0.88 (+0.47)

Target	Base rate	Vol-only	VIX-only	Price	Price + cross-asset
10% drawdown in next quarter	0.24	0.62 (+0.09)	0.61 (+0.08)	0.67 (+0.10)	0.64 (+0.07)
Negative next-quarter return	0.32	0.43 (−0.00)	0.39 (−0.00)	0.45 (−0.04)	0.44 (−0.08)
Bear, ±20% rule, ex post	0.19	0.71 (+0.15)	0.69 (+0.13)	0.83 (+0.27)	0.81 (+0.27)

The base rate is the share of test days in the bad class. For the volatility label it is 0.32, above the 17% of §6.2, because the threshold is an expanding quantile and the years since 2000 were more volatile than the history before them.

The 90% bootstrap intervals for the gain of the full model over the price-only model:

Target	Gain in AUC	Gain in Brier skill
Next-month volatility, top fifth	[+0.001, +0.025]	[−0.003, +0.039]
10% drawdown in next quarter	[−0.085, −0.006]	[−0.077, −0.003]
Negative next-quarter return	[−0.042, +0.028]	[−0.089, −0.007]
Bear, ±20% rule, ex post	[−0.036, +0.002]	[−0.030, +0.034]

Against the volatility-only model, the full model's gain on the volatility target is [+0.023, +0.056] in AUC and [+0.10, +0.17] in Brier skill.

Five findings:

1. **Volatility labels are learnable, and mostly from persistence.** Trailing realised volatility alone forecasts next month's volatility regime with an AUC of 0.84. The VIX does better (0.87), as a forward-looking input should, and the full model adds about another 0.01.
2. **Direction labels are not learnable.** Every input set scores below 0.5 on the negative-next-quarter label, and none beats climatology on the Brier score. An AUC below 0.5 means that the model's riskier days were followed by a loss *less* often than its safer days. What the models learned about the link between volatility and the next quarter's direction did not hold in the next period: 2009 and 2020 began their recoveries near the peak of measured risk.
3. **Drawdown labels sit in between, through volatility.** A 10% drawdown is more likely when volatility is high, simply because a large fall is easier when the market moves more. The skill is real but small (Brier skill about 0.1), and the cross-asset inputs make it worse rather than better.
4. **The ex-post bear label is nowcastable, but only as far as the drawdown has already happened.** Price features reach an AUC of 0.83. The model is largely reading the drawdown and the volatility that come with a bear market already under way (§6.5).
5. **The cross-asset inputs add almost nothing on these targets.** The VIX, the variance risk premium, the credit spread and the yield curve improve the volatility forecast over trailing volatility alone. Over the seven price features they add at most 0.01 of AUC on the volatility target (a 90% interval of +0.001 to +0.025, with the Brier-skill interval straddling zero), and on the other targets they make no difference or hurt. Two further checks point the same way. Adding the VIX/VIX3M term-structure ratio to models trained from September 2009, and testing on 2015–2025, left the volatility AUC at 0.77 and lowered the bear-label AUC from 0.78 to 0.76. A model *without* the term-structure ratio but trained from 1990 scored 0.80 and 0.81 on the same test years. Nineteen more years of history were worth more than the extra input. And with weak shrinkage ($\alpha = 0.001$), the twelve-input model's AUC on the drawdown label fell to 0.51, against 0.66 for the volatility-only model fitted the same way.

6.5 Detection at bear-market onsets

For the ex-post bear label, the question a practitioner asks is how soon after each peak the classifier says "bear". The table counts trading days from each peak since 2000 to the rule's confirmation, and to the first day the walk-forward classifier's probability reached one half.

Peak	Peak to trough (days)	Rule confirms	Price model	Price + cross-asset
24 Mar 2000	259	188	15	15

Peak	Peak to trough (days)	Rule confirms	Price model	Price + cross-asset
21 May 2001	82	78	4	6
19 Mar 2002	142	78	36	68
9 Oct 2007	283	192	69	31
6 Jan 2009	42	36	1	1
29 Apr 2011	108	108	67	68
20 Sep 2018	65	65	62	55
19 Feb 2020	23	16	12	6
8 Nov 2021	235	133	117	85

The classifiers cross one half well before the rule in the long bear markets of 2000 and 2007 (by 120 to 170 days), by 10 to 50 days in 2002, 2011 and 2021–22, and only days before it in the fast ones of 2018 and 2020. The first crossing flatters the classifier, though. The crossings of one to six days in 2001 and 2009 are artefacts: the probability had not fallen far below one half during the rally before each peak. Over the whole test period the full model's probability was above one half on 39% of ex-post bear days and on 6% of bull days, with 12 false alarms of five days or more; the confirmed state was on for 29% of bear days and 8% of bull days. The classifier is an earlier, noisier version of the rule.

Figure 2 shows why. The classifier's bear probability rises and falls with its forecast of volatility. It cannot see a bear-market rally such as November 2008 to January 2009, which the ±20% rule calls a bull market, and it lingers after the troughs, for weeks after March 2009 and less so after October 2022, when volatility was still elevated but the new bull market had begun.

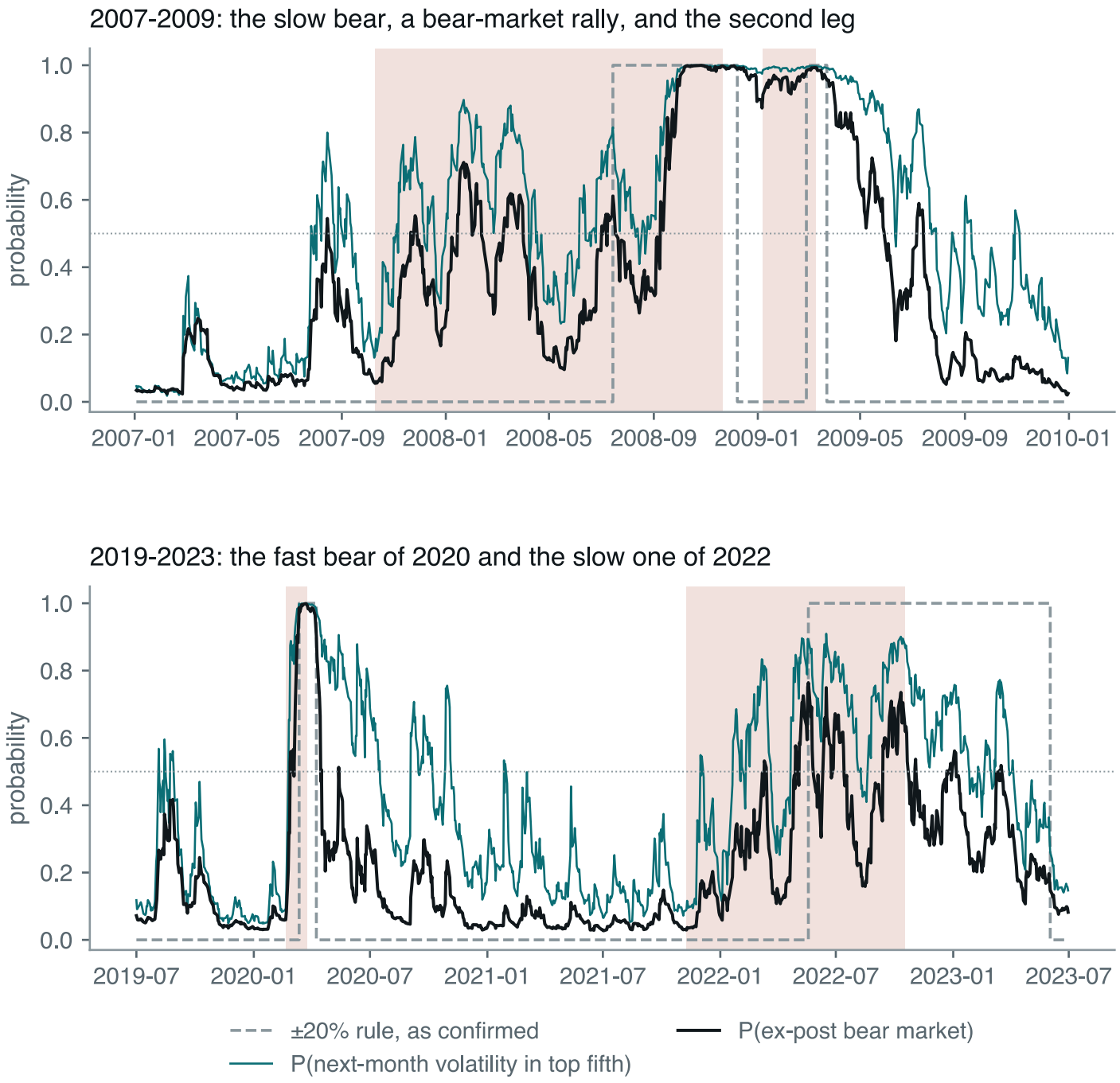


Figure 2. Walk-forward probabilities through the bear markets of 2007–09, 2020 and 2022. Shaded: ex-post ±20% bear markets. Black: the full model's probability of the ex-post bear label. Teal: its probability that next month's volatility lands in the top fifth. Dashed grey: the rule's confirmed state.

6.6 What the example does and does not show

The example supports the chapter's spine. Labels whose future part is volatility were forecast well, mostly from persistence; labels whose future part is direction were not forecast at all; the bull–bear label's value sat entirely in the days it could not be known; and the many inputs collapsed to the information already in the price path.

It does not show that cross-asset inputs are useless. Four limits matter:

1. **One market and one kind of label.** All the targets are about US equities. Labels about credit, about the stock–bond correlation, or about a multi-asset portfolio would give credit spreads and the curve a direct role, and the episodes of 2007 suggest credit can lead (§2.11).

2. **A linear model.** Interactions, such as high volatility together with widening spreads, were not offered. A tree ensemble could find them, at the cost of the few-episode problem.
3. **A short cross-asset history.** The cross-asset inputs enter in 1990, so the classifier's first refit, in January 2000, had seen two bear markets. The term-structure ratio starts in September 2009, so the first refit that used it, in January 2015, had seen one.
4. **Nine bear onsets.** Every statement about detection in §6.5 rests on nine episodes. The bootstrap intervals in §6.4 resample years, not episodes, and so understate the uncertainty about bear-market detection.

Within those limits, the practical reading is direct. Start a regime study with volatility labels and price-path inputs. Add the VIX. Add any other input only when a decision-specific label shows that it beats that baseline out of sample.

§6 Key takeaways

1. Eight reasonable definitions of a bad regime agree only moderately; apart from the ±20% and ±15% versions of the ex-post bear label, no pair reaches a kappa of 0.6.
2. The ±20% bull–bear label is worth a Sharpe ratio of 1.54 known in advance and 0.46 known when confirmed, against 0.45 for buy and hold. Its value lies in the 20% of days it cannot be known.
3. Next month's volatility regime is forecast with an AUC near 0.88; trailing volatility alone reaches 0.84.
4. Next quarter's direction is not forecast: every input set scored below 0.5 out of sample.
5. A bear-market classifier reads the drawdown and the volatility already present. It is an earlier, noisier version of the confirmation rule.
6. The VIX, credit spreads, the yield curve and the volatility term structure add at most 0.01 of AUC to price-path features for these equity labels; longer history was worth more than an extra input.

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7. Evaluating a regime classifier

A regime classifier produces a daily probability that is then turned into actions. Its evaluation therefore has three layers: the probability (is it accurate and calibrated?), the events (does it catch regime changes in time, and how often does it cry wolf?), and the decision (does acting on it pay after costs?). Each layer needs its own metric and the same discipline about uncertainty, because the sample is a few dozen episodes however many days it spans. The companion chapter's research protocol applies in full (section 11 of Market Regimes and Hidden Markov Models); this section adds what is specific to classifiers.

7.1 Proper scores and skill

A **proper scoring rule** is one whose expected value is best when the forecast probability equals the true probability, so it cannot be gamed by shading forecasts toward 0 or 1 (Gneiting & Raftery, 2007). Two are standard. The **Brier score** is the mean squared error of the probability over the n scored days,

$$\text{BS} = \frac{1}{n}\sum_{t=1}^{n}\left(\hat{p}_t - y_t\right)^2,$$

(Brier, 1950). The **log loss**, $-\frac{1}{n}\sum_t \left[y_t \log \hat{p}_t + (1-y_t)\log(1-\hat{p}_t)\right]$, punishes confident errors far more heavily. The Brier score is the better default for regime work, because a single confident miss in a crisis should count, but should not dominate.

Raw scores depend on the base rate, so report **skill** against a reference forecast:

$$\text{BSS} = 1 - \frac{\text{BS}_{\text{model}}}{\text{BS}_{\text{reference}}}.$$

A score of 0 means no better than the reference, 1 means perfect, and negative values mean worse. Use two references (§5.1): **climatology**, for comparison with other studies, and the **stale label**, for the decision. As reference points from §6.4, with the best input set for each label: the next-month volatility label reached a Brier skill of 0.47 over climatology, the drawdown label 0.10, and the direction label a negative skill.

7.2 Calibration

A probability is **calibrated** if, among the days on which it says 30%, the label turns out positive on about 30%. Check it with a reliability table: group forecasts into bins, and compare each bin's average forecast with its observed frequency. The Brier score splits into three parts, reliability minus resolution plus uncertainty (Murphy, 1973). With the forecasts grouped into bins k, bin k holding n_k days with mean forecast $\bar{f}_k$ and observed frequency $\bar{y}_k$, and with $\bar{y}$ the base rate over all n days,

$$\mathrm{BS} = \underbrace{\sum_k \tfrac{n_k}{n}\,(\bar{f}_k - \bar{y}_k)^2}_{\text{reliability}} - \underbrace{\sum_k \tfrac{n_k}{n}\,(\bar{y}_k - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}\,(1-\bar{y})}_{\text{uncertainty}}.$$

The identity is exact when all the forecasts in a bin are the same number; for continuous forecasts a small within-bin remainder is left over. *Reliability* measures miscalibration and should be small. *Resolution* measures how far the bins' observed frequencies depart from the base rate, that is, how well the forecasts separate good days from bad, and should be large. *Uncertainty* is the base rate's own variance, which no model can change.

Calibrate on training data only (on out-of-fold scores for a flexible model, §5.11), refit the calibration with the model, and check calibration separately in calm and stressed periods: a model can be calibrated on average and badly miscalibrated in exactly the periods that matter.

7.3 Discrimination when the class is rare

The **AUC** is the area under the receiver operating characteristic curve, which plots the share of positive days flagged against the share of negative days flagged as the threshold moves. It equals the probability that a randomly chosen positive day receives a higher score than a randomly chosen negative day. It ignores calibration entirely, since any monotone transformation of the scores leaves it unchanged, so it measures ranking and nothing else. For a rough sense of scale, suppose the scores of positive and negative days were normal with equal spread and means Δ standard deviations apart. A positive score minus a negative score is then normal with mean Δ and variance 2, so AUC $= \Phi(\Delta/\sqrt{2})$, with Φ the standard normal distribution function. An AUC of 0.84 means Δ is about 1.4, and an AUC of 0.6 means about 0.36.

When the class of interest is rare, as crisis regimes are, the AUC can look reassuring while most of the alarms are false. The **precision–recall curve**, which plots the share of alarms that are correct against the share of positives caught, is more informative in that case (Saito & Rehmsmeier, 2015). Report precision at the threshold the decision will actually use.

7.4 Event metrics: delay, false alarms and flicker

Scores averaged over days hide what a regime classifier is for: catching changes. Four event metrics, computed per episode:

1. **Detection delay.** Days from the start of each positive episode to the first day the probability crosses the decision threshold. Report the median and the worst case, and compare them with the stale label's delay. For the ex-post bear label in §6.5, the price-feature classifier's delay ranged from 1 to 117 trading days (the shortest are crossings that never reset after the previous bear market, as in 2009), against 16 to 192 for the confirmation rule.
2. **False alarms.** Runs of at least m days above the threshold while the label is negative, per year. Set m by the decision: a sizing rule may tolerate one-day blips that a strategy switch cannot.
3. **Missed episodes.** Positive episodes never detected before they end.
4. **Flips per year.** The number of regime changes the post-processed output makes. Each flip has a trading cost, and the companion chapter's cube-root rule for hysteresis bands applies (section 12 of Market Regimes and Hidden Markov Models).

Delay and false alarms trade against each other as the threshold moves. Plot the whole trade-off rather than one point on it. This is the event-level analogue of the receiver operating characteristic, and the literature on monitoring for changes in behaviour treats it the same way (Fawcett & Provost, 1999).

7.5 Economic evaluation

The final test is the decision itself: what does acting on the classifier earn, after costs, against two alternatives? The first is the stale-label rule, which needs no model. The second is the continuous forecast that the regime coarsens (§5.10), such as sizing inversely to forecast volatility without any regime. A regime classifier that beats neither is not worth its complexity.

In the worked example the real-time ±20% rule, used for timing, earned what buy and hold earned (§6.3); the classifier's own timing value would have to be measured the same way, with its false alarms paid for: map the probability to a position by a stated rule, apply it with a one-day lag, charge a cost for every change of position, and compare the Sharpe ratio and maximum drawdown with those of the two alternatives.

7.6 Uncertainty: count episodes, not days

Three tools, in increasing order of strength:

1. **Block bootstrap.** Resample blocks of the test period long enough to contain whole episodes, a year or more for bear markets, as in §6.4 or, with random block lengths, as in the stationary bootstrap (Politis & Romano, 1994). Intervals from day-level resampling are far too narrow.
2. **A persistence-preserving permutation test.** Break the link between the features and the label while keeping each one's persistence: circularly shift the label sequence against the features by random offsets of at least a year, refit, and score. The distribution of scores under this null shows how much skill persistence and overlap alone produce.
3. **A synthetic null.** Simulate price paths from a model with no regimes, such as a GARCH process fitted to the same data, compute the same labels, and run the whole pipeline. Any "skill" it finds in direction labels is an artefact of the pipeline.

Count the trials. Every label definition, threshold, horizon and input set tried is a trial, and the best of many trials overstates skill (Bailey & López de Prado, 2014; Harvey, Liu & Zhu, 2016).

7.7 A protocol

1. Write the decision, label, horizon and reach in one sentence (§1.2).
2. Vet the label: decision relevance, episodes, stability, reach, and the oracle and stale-label values (§4.9).
3. Build point-in-time inputs and a walk-forward training set that uses only labels final at each refit date (§4.10).
4. Fit the stale-label baseline and a regularised logistic model before anything else (§5.1, §5.4).
5. Score probabilities with the Brier score against climatology and against the stale label, check calibration, and report AUC with precision at the decision threshold (§7.1–§7.3).
6. Score events: delay, false alarms, misses and flips, per episode (§7.4).
7. Score the decision after costs against the stale-label rule and the continuous alternative (§7.5).
8. Put intervals on every comparison with a block bootstrap, run the permutation and synthetic nulls, and count the trials (§7.6).

§7 Key takeaways

1. Score probabilities with a proper rule, the Brier score by default, as skill over climatology and over the stale label.
2. Check calibration in calm and stressed periods separately.
3. AUC measures ranking only. For rare crisis classes, report precision at the threshold you will use.
4. Score events, not just days: detection delay, false alarms, misses and flips, per episode.
5. The decision test compares the classifier with the stale-label rule and with the continuous forecast it coarsens, after costs.
6. Resample whole episodes, test against nulls that keep persistence, and count every label, threshold and input set tried.

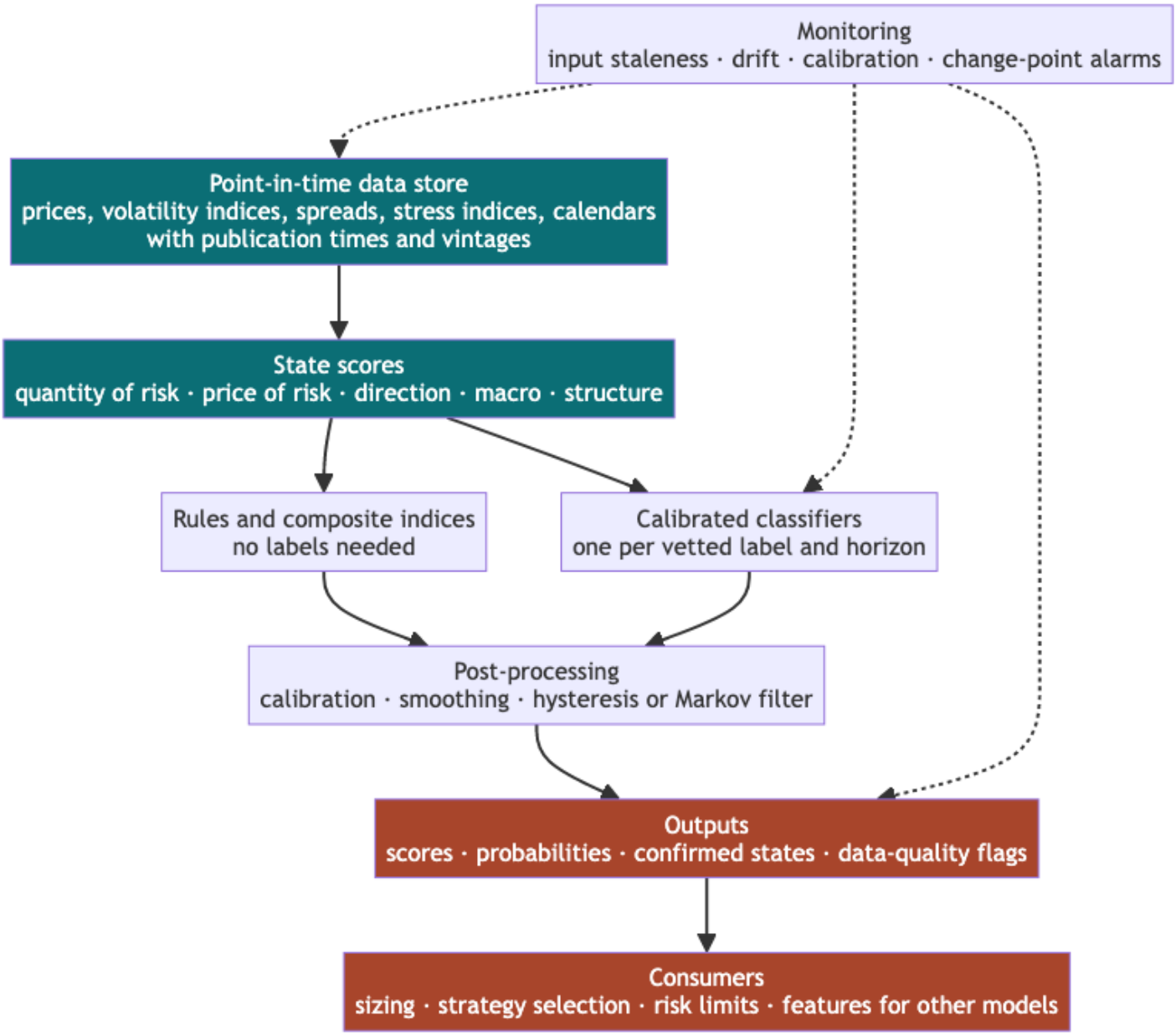
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8. Building a regime study

This section turns the chapter into an architecture. It is generic: it assumes a daily data store, a study framework that computes derived series from inputs, and consumers such as position sizing. Its main recommendation is to make the study produce continuous scores and calibrated probabilities, with discrete regimes derived from them, rather than a single regime label.

8.1 A reference architecture

The study has five layers. Data flow through them in one direction, and a monitoring layer watches the data, the classifiers and the outputs.



The data store holds every input with the time it became available and, for revised series, every vintage (§3.11). Studies read only what was available at each date.

The state scores are a handful of continuous, causally standardised composites, one or two per state variable (§3.13): for example, a quantity-of-risk score from realised volatility at several horizons and the VIX, and a price-of-risk score from credit spreads and the variance risk premium. They are useful outputs in their own right.

The regime models are of two kinds. Rules and composite indices need no labels and produce regimes with zero reach. Classifiers are trained, one per vetted label and horizon, and produce probabilities.

Post-processing calibrates the probabilities and smooths them into regimes (§5.11).

The outputs go to consumers, each of which uses the output that suits it: sizing uses the risk score or the volatility probability, a strategy switch uses a smoothed regime, a risk limit uses a rule.

8.2 What the study should output

For each date, a regime study should publish:

1. **The state scores**, as continuous values.
2. **A calibrated probability for each label**, tagged with the label's definition and horizon. "P = 0.7" is meaningless without "that next month's realised volatility exceeds its historical 80th percentile".
3. **The stale label or confirmed state**, so that consumers and monitors can see the baseline the model is supposed to beat.
4. **The discrete regime**, derived from the probability by a stated rule, never as the only output.
5. **Data-quality flags**: which inputs were stale, missing or filled.
6. **The model version and training window**, so that any output can be reproduced.

A study that outputs only a regime label throws away the probability, the uncertainty and the reason, and its consumers cannot size by confidence.

8.3 A staged build with gates

The early stages are infrastructure, and skipping them is the usual reason regime studies fail in production. Each stage has a gate that must be passed before the next begins.

Stage	Build	Gate
0. Data	Point-in-time store; publication lags; vintages for revised series	An audit shows no input is used before its publication time
1. Scores and rules	State scores; rule-based regimes with hysteresis; stale labels	Scores and rules behave as expected in known episodes (2008, 2020, 2022)
2. Labels	Candidate labels for the decision; the five vetting checks	At least one label passes all five checks (§4.9)
3. Classifier	Regularised logistic model on five to ten inputs, walk-forward	Brier skill over the stale label, with a block-bootstrap interval above zero
4. Decision	The decision rule using the calibrated probability	Beats the stale-label rule and the continuous alternative after costs
5. Production	Monitoring, retraining schedule, fallbacks	Drift and staleness alarms tested by injecting stale data

Most studies should stop at stage 1 or 2. A set of well-built state scores and rules, with hysteresis, delivers most of what a regime study can deliver for risk decisions, and stage 3 is worth building only where a vetted label shows a gap that a model can close.

8.4 Failure modes in production

- **Stale or missing inputs.** A source stops updating, or a holiday calendar differs between sources, and the study silently forward-fills an old value. Flag inputs older than their expected publication lag, and degrade to rules that do not use them.
- **Revisions and backfills.** A vendor revises history, and the next retraining learns from data that were never available live. Store what was fetched, when it was fetched.
- **Structural breaks in an input.** The composition of option volume changed the meaning of put/call ratios (§3.9); an index methodology change shifts a level. Monitor each input's distribution against its own history.
- **Label drift.** The base rate of a label changes, for example because an expanding threshold adapts only slowly: the volatility label of §6.2 marks 17% of days over the century but 32% of the days since 2000 (§6.4). Report the base rate in each training window.
- **Calibration decay.** Probabilities that were calibrated in training drift. Recheck the reliability table every retraining.
- **Regimes that change meaning.** A retrained latent-state model or jump model swaps or redefines its states. Identify states by a fixed property and alarm on large label revisions.
- **Crowding.** Many funds cutting risk on the same VIX threshold may amplify the moves they respond to **[Hypothesis]**. Prefer continuous sizing to hard thresholds, which also reduces the study's own trading.

§8 Key takeaways

1. Build in layers: point-in-time data, continuous state scores, rules and classifiers, post-processing, outputs, with monitoring across all of them.
2. Output scores and calibrated probabilities with their label definitions, the stale baseline, data-quality flags and versions. Never only a regime label.
3. Pass the gates in order. Data integrity and well-behaved state scores come before any classifier.
4. Most studies should stop at state scores and rules with hysteresis. Train a classifier only where a vetted label shows a gap a model can close.
5. In production, the failures are stale inputs, revisions, structural breaks in inputs and decaying calibration. Monitor all four.

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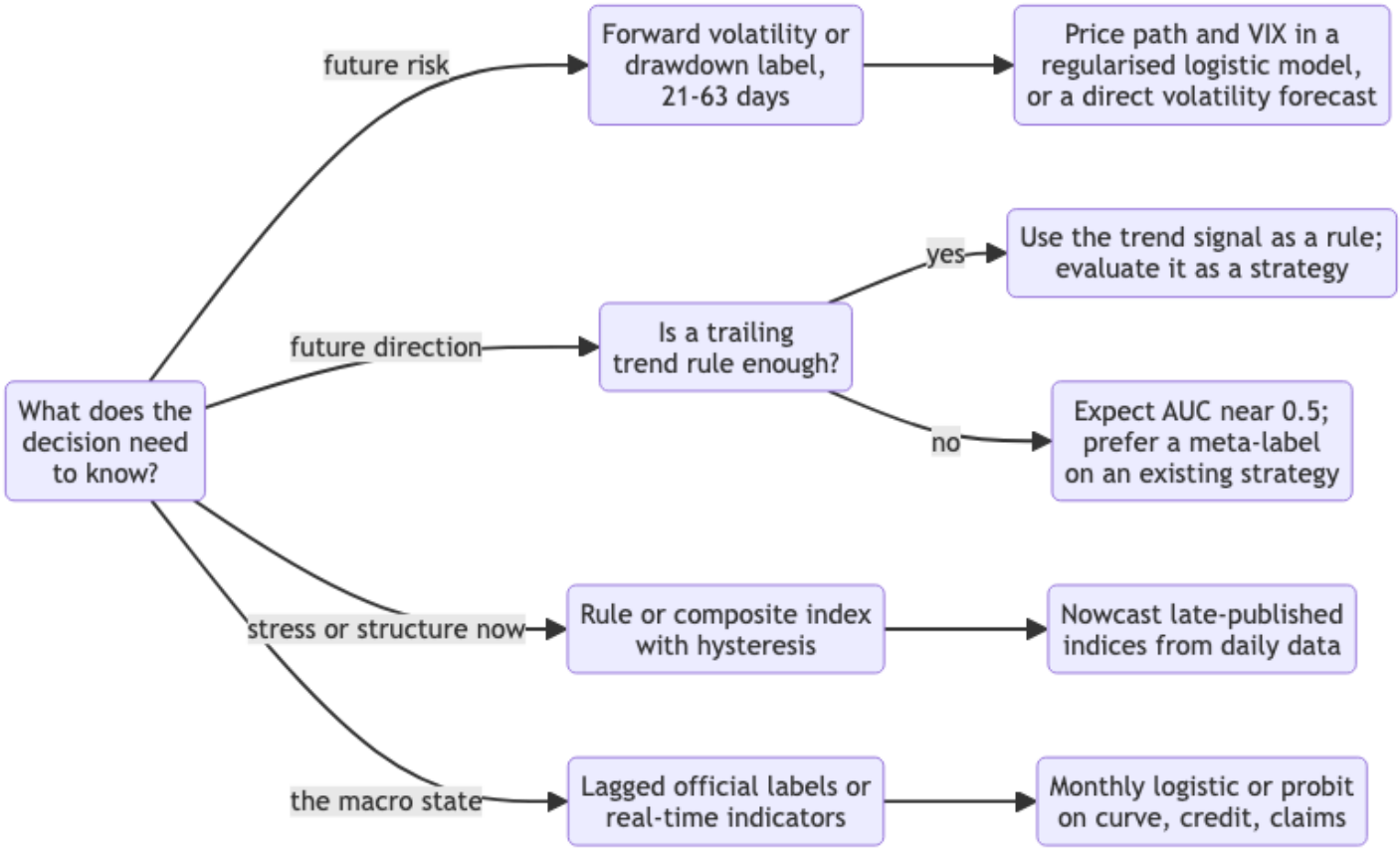
9. Synthesis

9.1 The framework in one paragraph

A regime is a coarse partition of a few continuous state variables, chosen to serve a decision: the quantity of risk, the price of risk, direction, the macroeconomic state and structure. Inputs are noisy measurements of those state variables, and most of them measure the quantity of risk. A label is a rule on a window of the price path, and its reach (how far into the future the window extends) decides everything else: the purge in training, the floor on detection delay, and how much a classifier can learn. A supervised regime classifier is a forecaster of its label's future part. When that future part is volatility, which persists, the classifier learns a great deal, mostly from the stale label. When it is direction, which does not persist, the classifier learns almost nothing, and the label's value sits in the days when it cannot be known. The sample is the number of episodes, so models must be small and evaluation must resample episodes.

Decision	State variable	Label	Reach	First inputs	First model	Baseline to beat
Sizing, risk limits	quantity of risk	forward volatility quantile	21 days	realised volatility, VIX	logistic, or a direct volatility forecast	trailing volatility
Drawdown control	quantity of risk, direction	forward drawdown event	63 days	realised volatility, drawdown, VIX	logistic	trailing volatility
Exposure timing	direction	ex-post bull–bear or forward return	months	price path	none worth fitting	confirmed state; buy and hold
Leverage, liquidity	price of risk, structure	stress index above a threshold	0, plus publication lag	spreads, stress indices	rule with hysteresis; nowcast of the index	the last published value
Strategic allocation	macro	lagged recession or growth–inflation phase	months	curve, credit, claims, surprises	monthly logistic or probit	the last published phase
Diversification	structure	correlation regime	0 (estimated)	rolling correlations, implied correlation	rule with hysteresis	rolling estimate
Strategy selection	depends	meta-label of the base strategy	its holding period	the strategy's own state plus risk inputs	logistic	always on

9.2 Decision tree



Whichever branch applies, the same steps come first: write the decision as a sentence, vet the label (§4.9), build the stale-label baseline, align every input to its publication time, and count episodes rather than days.

9.3 Ten rules for someone starting today

1. Write the decision, the label, the horizon and the reach in one sentence before loading any data.
2. Prefer labels whose future part is volatility. They can be forecast; direction cannot.
3. Compute the label's reach and purge accordingly. A label with zero reach is a rule: compute it, do not learn it.
4. Beat the stale label, not the base rate.
5. Start with the price path and the VIX. Add any other input only when it beats that baseline out of sample on a decision-specific label.
6. Align every input to its publication time and store vintages of revised series.
7. Use expanding percentiles for levels; keep rolling z-scores for surprises.
8. Use a regularised logistic model on five to ten inputs before anything more complex, and calibrate whatever you use.
9. Output continuous scores and calibrated probabilities; derive regimes from them with hysteresis.
10. Count episodes, not days: resample them, test against nulls that keep persistence, and count every trial.

9.4 What is known, and what is not

Established. Volatility is persistent and forecastable from its own history and from implied volatility, so regimes defined by future risk can be forecast with real skill **[Fact]**. Official macroeconomic regime labels arrive months to years late and are not observable in real time **[Fact]**.

Contested. Whether regimes defined by future direction can be forecast with useful skill from public market data. Most proposed predictors of the equity premium fail out of sample (Welch & Goyal, 2008), and the worked example finds the same for regime labels. Forecasts with sensible restrictions (Campbell & Thompson, 2008) or combinations of many predictors (Rapach, Strauss & Zhou, 2010) recover small but real gains. The disagreement is about whether small means useful; no one finds the large skill a direction regime would need. Whether cross-asset inputs (credit, the curve, the volatility term structure, sentiment) add to price-path inputs for equity regimes. The worked example, which did not test sentiment, finds a gain of at most 0.01 of AUC for its labels; practitioners report gains in credit-led episodes such as 2007, and teacher–student

studies report gains for multi-asset allocation. Both sides rest on few episodes. Whether volatility timing improves returns, rather than only risk, out of sample (Moreira & Muir, 2017; Cederburg et al., 2020).

Open. Whether labels defined directly on multi-asset outcomes, such as the stock–bond correlation or a stress event, are forecastable from structure inputs with useful skill. How to share episodes across markets, so that a classifier trained on many countries' bear markets is less starved of data than one trained on the US alone. And whether the composition changes in option markets since 2020 have created new information in sentiment inputs or destroyed the old.

This chapter's view, stated as such: a regime study earns its keep as a disciplined measurement of the quantity and price of risk, with probabilities for a few risk-type labels on top. The search for a classifier that tells bull markets from bear markets early is the search for a return forecast, and should be judged as one.

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10. References

Grouped by kind. Each entry carries one line on why it matters. Practitioner sources name the author's interest; critiques have their own group. Links go to a free copy where one exists; entries marked paywalled link to the publisher.

10.1 Regime definitions and dating

- **Hamilton, W. P. (1922).** *The Stock Market Barometer.* Harper & Brothers. — Dow theory's primary bull and bear markets, confirmed by two averages: the origin of the lagging bull–bear definition. No stable free copy located.
- **Burns, A. F. & Mitchell, W. C. (1946).** *Measuring Business Cycles.* NBER. — The reference-cycle method behind official recession dates.
- **Bry, G. & Boschan, C. (1971).** *Cyclical Analysis of Time Series: Selected Procedures and Computer Programs.* NBER Technical Paper 20. — The turning-point algorithm that made dating reproducible, still used by the OECD.
- **Hamilton, J. D. (1989).** "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle." *Econometrica* 57(2), 357–384. — The regime as a latent Markov state; the companion chapter's subject.
- **Harding, D. & Pagan, A. (2002).** "Dissecting the Cycle: A Methodological Investigation." *Journal of Monetary Economics* 49(2), 365–381. [publisher — paywalled] — The general framework for describing cycles by their turning points.
- **Pagan, A. R. & Sossounov, K. A. (2003).** "A Simple Framework for Analysing Bull and Bear Markets." *Journal of Applied Econometrics* 18(1), 23–46. [DOI] — Bry–Boschan adapted to monthly stock prices, with minimum phase and cycle lengths.
- **Lunde, A. & Timmermann, A. (2004).** "Duration Dependence in Stock Prices: An Analysis of Bull and Bear Markets." *Journal of Business & Economic Statistics* 22(3), 253–273. [publisher — paywalled] — The threshold rule with separate up and down thresholds, and the duration dependence of the phases it dates.
- **Gonzalez, L., Powell, J. G., Shi, J. & Wilson, A. (2005).** "Two Centuries of Bull and Bear Market Cycles." *International Review of Economics & Finance* 14(4), 469–486. [publisher — paywalled] — Dating over the longest US history, for more episodes.
- **Maheu, J. M., McCurdy, T. H. & Song, Y. (2012).** "Components of Bull and Bear Markets: Bull Corrections and Bear Rallies." *Journal of Business & Economic Statistics* 30(3), 391–403. — Four states, separating corrections and rallies from the phases they interrupt.
- **Kole, E. & van Dijk, D. (2017).** "How to Identify and Forecast Bull and Bear Markets?" *Journal of Applied Econometrics* 32(1), 120–139. [publisher — paywalled] — Rules date better in sample; switching models forecast better.
- **Liu, J., Maheu, J. M. & Song, Y. (2024).** "Identification and Forecasting of Bull and Bear Markets Using Multivariate Returns." *Journal of Applied Econometrics* 39(5), 723–745. [publisher — paywalled] — Several markets' returns jointly sharpen both dating and forecasting.
- **Cooper, M. J., Gutierrez, R. C. & Hameed, A. (2004).** "Market States and Momentum." *Journal of Finance* 59(3), 1345–1365. — The UP/DOWN market state from the trailing 36-month return, and momentum's dependence on it.
- **Daniel, K. & Moskowitz, T. J. (2016).** "Momentum Crashes." *Journal of Financial Economics* 122(2), 221–247. — The bear-state indicator, and why momentum crashes when the market rebounds from it.
- **Goulding, C. L., Harvey, C. R. & Mazzoleni, M. G. (2023).** "Momentum Turning Points." *Journal of Financial Economics* 149(3), 378–406. — Bull, Correction, Bear and Rebound from slow and fast momentum signs.
- **Lo, A. W. & MacKinlay, A. C. (1988).** "Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test." *Review of Financial Studies* 1(1), 41–66. — The variance ratio, the standard measure behind trending and mean-reverting regimes.
- **Chari, A., Dilts Stedman, K. & Lundblad, C. (2023).** "Risk-On Risk-Off: A Multifaceted Approach to Measuring Global Investor Risk Aversion." NBER Working Paper 31907. — A risk-on/risk-off index from several market measures.

- **Miranda-Agrippino, S. & Rey, H. (2020).** "U.S. Monetary Policy and the Global Financial Cycle." *Review of Economic Studies* 87(6), 2754–2776. — One global factor behind risky-asset prices, moved by US monetary policy.
- **Bekaert, G., Hoerova, M. & Lo Duca, M. (2013).** "Risk, Uncertainty and Monetary Policy." *Journal of Monetary Economics* 60(7), 771–788. — The VIX split into risk aversion and uncertainty.
- **Chauvet, M. & Piger, J. (2008).** "A Comparison of the Real-Time Performance of Business Cycle Dating Methods." *Journal of Business & Economic Statistics* 26(1), 42–49. — Turning points called in real time, against the NBER's later dates.
- **Hamilton, J. D. (2011).** "Calling Recessions in Real Time." *International Journal of Forecasting* 27(4), 1006–1026. — A survey of real-time recession dating and its record.
- **Giusto, A. & Piger, J. (2017).** "Identifying Business Cycle Turning Points in Real Time with Vector Quantization." *International Journal of Forecasting* 33(1), 174–184. [publisher — paywalled] — A simple classifier for real-time turning points.
- **Sahm, C. (2019).** "Direct Stimulus Payments to Individuals." In *Recession Ready*, The Hamilton Project and the Washington Center for Equitable Growth. — The Sahm rule, a real-time recession signal from unemployment.
- **NBER (various years).** Business Cycle Dating Committee announcements. — The dates of each announcement, from which the label lags of §2.8 follow.
- **OECD (2012).** *OECD System of Composite Leading Indicators.* Prepared by G. Gyomai and E. Guidetti. — How the composite leading indicators are built, including the Hodrick–Prescott detrending and Bry–Boschan dating.
- **Guidolin, M. & Timmermann, A. (2007).** "Asset Allocation under Multivariate Regime Switching." *Journal of Economic Dynamics and Control* 31(11), 3503–3544. — Four joint states for stocks and bonds: crash, slow growth, bull, recovery.
- **Ang, A. & Timmermann, A. (2012).** "Regime Changes and Financial Markets." *Annual Review of Financial Economics* 4, 313–337. — The best single review of regime models in finance.
- **Campbell, J. Y., Sunderam, A. & Viceira, L. M. (2017).** "Inflation Bets or Deflation Hedges? The Changing Risks of Nominal Bonds." *Critical Finance Review* 6(2), 263–301. — Why the stock–bond correlation changes sign.
- **Baele, L., Bekaert, G. & Inghelbrecht, K. (2010).** "The Determinants of Stock and Bond Return Comovements." *Review of Financial Studies* 23(6), 2374–2428. — Macroeconomic fundamentals explain little of the stock–bond correlation; liquidity factors more.
- **Pollet, J. M. & Wilson, M. (2010).** "Average Correlation and Stock Market Returns." *Journal of Financial Economics* 96(3), 364–380. — Average correlation, not average variance, predicts market returns.
- **Driessen, J., Maenhout, P. J. & Vilkov, G. (2009).** "The Price of Correlation Risk: Evidence from Equity Options." *Journal of Finance* 64(3), 1377–1406. — Correlation risk is priced in index options; the basis of implied-correlation inputs.
- **Stivers, C. & Sun, L. (2010).** "Cross-Sectional Return Dispersion and Time Variation in Value and Momentum Premiums." *Journal of Financial and Quantitative Analysis* 45(4), 987–1014. [publisher — paywalled] — Dispersion as a state variable for factor premiums.

10.2 Inputs

- **Andersen, T. G., Bollerslev, T., Diebold, F. X. & Labys, P. (2003).** "Modeling and Forecasting Realized Volatility." *Econometrica* 71(2), 579–625. — Realised variance from intraday returns as a precise volatility measurement.
- **Corsi, F. (2009).** "A Simple Approximate Long-Memory Model of Realized Volatility." *Journal of Financial Econometrics* 7(2), 174–196. [publisher — paywalled] — The HAR model, the standard direct forecast of volatility.
- **Moskowitz, T. J., Ooi, Y. H. & Pedersen, L. H. (2012).** "Time Series Momentum." *Journal of Financial Economics* 104(2), 228–250. [publisher — paywalled] — What trailing returns forecast, across asset classes.
- **Cboe (current).** *Cboe Volatility Index Methodology.* — How the VIX is computed from S&P 500 option prices.
- **Johnson, T. L. (2017).** "Risk Premia and the VIX Term Structure." *Journal of Financial and Quantitative Analysis* 52(6), 2461–2490. — The slope of the VIX term structure and the returns of volatility assets.
- **Simon, D. P. & Campasano, J. (2014).** "The VIX Futures Basis: Evidence and Trading Strategies." *Journal of Derivatives* 21(3), 54–69. — What the futures term structure says about subsequent VIX futures returns.
- **Bollerslev, T., Tauchen, G. & Zhou, H. (2009).** "Expected Stock Returns and Variance Risk Premia." *Review of Financial Studies* 22(11), 4463–4492. [publisher — paywalled] — The variance risk premium as a predictor of equity returns.
- **Bekaert, G. & Hoerova, M. (2014).** "The VIX, the Variance Premium and Stock Market Volatility." *Journal of Econometrics* 183(2), 181–192. — The VIX split into expected variance and the premium, with different predictive roles.
- **Gilchrist, S. & Zakrajšek, E. (2012).** "Credit Spreads and Business Cycle Fluctuations." *American Economic Review* 102(4), 1692–1720. — The excess bond premium.
- **Favara, G., Gilchrist, S., Lewis, K. F. & Zakrajšek, E. (2016).** "Updating the Recession Risk and the Excess Bond Premium." FEDS Notes. — The Federal Reserve's maintained version of the series.
- **Estrella, A. & Hardouvelis, G. A. (1991).** "The Term Structure as a Predictor of Real Economic Activity." *Journal of Finance* 46(2), 555–576. [publisher — paywalled] — The yield curve as a leading indicator.

- **Estrella, A. & Mishkin, F. S. (1998).** "Predicting U.S. Recessions: Financial Variables as Leading Indicators." *Review of Economics and Statistics* 80(1), 45–61. — The term spread beats other financial variables beyond one quarter.
- **Engstrom, E. C. & Sharpe, S. A. (2019).** "The Near-Term Forward Yield Spread as a Leading Indicator: A Less Distorted Mirror." *Financial Analysts Journal* 75(4), 37–49. — A cleaner curve signal than long-minus-short spreads.
- **Liu, W. & Moench, E. (2016).** "What Predicts US Recessions?" *International Journal of Forecasting* 32(4), 1138–1150. — A systematic comparison of leading indicators.
- **Kliesen, K. L. & Smith, D. C. (2010).** "Measuring Financial Market Stress." *Economic Synopses* 2010(2), Federal Reserve Bank of St. Louis. — The St. Louis Fed stress index.
- **Brave, S. & Butters, R. A. (2012).** "Diagnosing the Financial System: Financial Conditions and Financial Stress." *International Journal of Central Banking* 8(2), 191–239. — The NFCI, and its record in predicting stress.
- **Hollo, D., Kremer, M. & Lo Duca, M. (2012).** "CISS — A Composite Indicator of Systemic Stress in the Financial System." ECB Working Paper 1426. — Aggregation that weights stress by how systemic it is.
- **Monin, P. J. (2019).** "The OFR Financial Stress Index." *Risks* 7(1), 25. — A daily, unrevised stress index from 33 market variables.
- **Kritzman, M. & Li, Y. (2010).** "Skulls, Financial Turbulence, and Risk Management." *Financial Analysts Journal* 66(5), 30–41. [publisher — paywalled] — The turbulence index. **[Contested]**: the authors' firm sells analytics built on it.
- **Kritzman, M., Li, Y., Page, S. & Rigobon, R. (2011).** "Principal Components as a Measure of Systemic Risk." *Journal of Portfolio Management* 37(4), 112–126. — The absorption ratio. Same interest as above.
- **Adrian, T., Boyarchenko, N. & Giannone, D. (2019).** "Vulnerable Growth." *American Economic Review* 109(4), 1263–1289. — Financial conditions forecast the lower tail of growth, not its median.
- **Baker, M. & Wurgler, J. (2006).** "Investor Sentiment and the Cross-Section of Stock Returns." *Journal of Finance* 61(4), 1645–1680. — The composite sentiment index.
- **Simon, D. P. & Wiggins, R. A. (2001).** "S&P Futures Returns and Contrary Sentiment Indicators." *Journal of Futures Markets* 21(5), 447–462. [publisher — paywalled] — The VIX, put/call and TRIN as contrarian signals.
- **Pan, J. & Poteshman, A. M. (2006).** "The Information in Option Volume for Future Stock Prices." *Review of Financial Studies* 19(3), 871–908. — Option volume predicts returns through non-public open-buy volume, not the public ratios.
- **Bryzgalova, S., Pavlova, A. & Sikorskaya, T. (2023).** "Retail Trading in Options and the Rise of the Big Three Wholesalers." *Journal of Finance* 78(6), 3465–3514. [publisher — paywalled] — The retail option boom that changed what option-volume ratios measure.
- **Cboe (2025).** "SPX 0DTE Options Jump to 61% Share on Retail Resurgence." Cboe Insights. — The share of same-day expiries in S&P 500 option volume. The exchange profits from this volume.
- **Aruoba, S. B., Diebold, F. X. & Scotti, C. (2009).** "Real-Time Measurement of Business Conditions." *Journal of Business & Economic Statistics* 27(4), 417–427. — A daily business-conditions index from mixed-frequency data.
- **Scotti, C. (2016).** "Surprise and Uncertainty Indexes: Real-Time Aggregation of Real-Activity Macro-Surprises." *Journal of Monetary Economics* 82, 1–19. [publisher — paywalled] — How to build a macroeconomic surprise index.
- **Croushore, D. & Stark, T. (2001).** "A Real-Time Data Set for Macroeconomists." *Journal of Econometrics* 105(1), 111–130. [publisher — paywalled] — Why vintages matter, and the data set that makes them available.
- **Giannone, D., Reichlin, L. & Small, D. (2008).** "Nowcasting: The Real-Time Informational Content of Macroeconomic Data." *Journal of Monetary Economics* 55(4), 665–676. [publisher — paywalled] — Updating an estimate of the present as each release arrives.

10.3 Labels, classifiers and machine learning

- **López de Prado, M. (2018).** *Advances in Financial Machine Learning.* Wiley. — Triple-barrier labels, meta-labels, sample uniqueness, and purged and embargoed cross-validation.
- **López de Prado, M. (2020).** *Machine Learning for Asset Managers.* Cambridge Elements in Quantitative Finance. — Trend-scanning labels (chapter 5).
- **Glattfelder, J. B., Dupuis, A. & Olsen, R. B. (2011).** "Patterns in High-Frequency FX Data: Discovery of 12 Empirical Scaling Laws." *Quantitative Finance* 11(4), 599–614. [publisher — paywalled] — Directional-change event series.
- **Tsang, E. P. K., Tao, R., Serguieva, A. & Ma, S. (2017).** "Profiling High-Frequency Equity Price Movements in Directional Changes." *Quantitative Finance* 17(2), 217–225. [publisher — paywalled] — Directional changes as regime descriptors.
- **Kim, C.-J. (1994).** "Dynamic Linear Models with Markov-Switching." *Journal of Econometrics* 60(1–2), 1–22. [publisher — paywalled] — The smoother whose output becomes a soft label, and a leak when used as a signal.
- **Bemporad, A., Breschi, V., Piga, D. & Boyd, S. (2018).** "Fitting Jump Models." *Automatica* 96, 11–21. — The jump model and its fitting algorithm.
- **Nystrup, P., Lindström, E. & Madsen, H. (2020).** "Learning Hidden Markov Models with Persistent States by Penalizing Jumps." *Expert Systems with Applications* 150, 113307. [publisher — paywalled] — Jump models for financial regimes.
- **Aydınhan, A. O., Kolm, P. N., Mulvey, J. M. & Shu, Y. (2024).** "Identifying Patterns in Financial Markets: Extending the Statistical Jump Model for Regime Identification." *Annals of Operations Research*. [DOI] — Continuous and

probabilistic extensions of the jump model.
- **Horvath, B., Issa, Z. & Muguruza, A. (2024).** "Clustering Market Regimes Using the Wasserstein Distance." *Journal of Computational Finance*. — Clustering whole return distributions rather than feature vectors.
- **Botte, A. & Bao, D. (2021).** "A Machine Learning Approach to Regime Modeling." Two Sigma. — A Gaussian mixture on factor returns with four named conditions. The authors' firm sells factor analytics.
- **Shu, Y., Yu, C. & Mulvey, J. M. (2024a).** "Downside Risk Reduction Using Regime-Switching Signals: A Statistical Jump Model Approach." *Journal of Asset Management* 25(5), 493–507. — Jump-model regimes for drawdown control.
- **Shu, Y., Yu, C. & Mulvey, J. M. (2024b).** "Dynamic Asset Allocation with Asset-Specific Regime Forecasts." *Annals of Operations Research* 346(1), 285–318. — The teacher–student design: jump-model labels forecast by gradient-boosted trees.
- **Shu, Y. & Mulvey, J. M. (2024).** "Dynamic Factor Allocation Leveraging Regime-Switching Signals." *Journal of Portfolio Management* 51(3), 50–72. — The same design for factor allocation.
- **Hauptmann, J., Hoppenkamps, A., Min, A., Ramsauer, F. & Zagst, R. (2014).** "Forecasting Market Turbulence Using Regime-Switching Models." *Financial Markets and Portfolio Management* 28(2), 139–164. [publisher — paywalled] — An early teacher–student system: logistic regressions forecasting Markov-switching probabilities.
- **Uysal, A. S. & Mulvey, J. M. (2021).** "A Machine Learning Approach in Regime-Switching Risk Parity Portfolios." *Journal of Financial Data Science* 3(2), 87–108. [publisher — paywalled] — Random forests on macro data for recession and contraction probabilities.
- **Kauppi, H. & Saikkonen, P. (2008).** "Predicting U.S. Recessions with Dynamic Binary Response Models." *Review of Economics and Statistics* 90(4), 777–791. [publisher — paywalled] — Dynamic probit models for recessions.
- **Nyberg, H. (2013).** "Predicting Bear and Bull Stock Markets with Dynamic Binary Time Series Models." *Journal of Banking & Finance* 37(9), 3351–3363. [publisher — paywalled] — The same models for bull and bear markets.
- **Chen, S.-S. (2009).** "Predicting the Bear Stock Market: Macroeconomic Variables as Leading Indicators." *Journal of Banking & Finance* 33(2), 211–223. [publisher — paywalled] — Which macro variables help predict bear markets.
- **Diebold, F. X., Lee, J.-H. & Weinbach, G. C. (1994).** "Regime Switching with Time-Varying Transition Probabilities." In C. P. Hargreaves, ed., *Nonstationary Time Series Analysis and Cointegration*, Oxford University Press, 283–302. [publisher — paywalled] — Transitions driven by observables.
- **Filardo, A. J. (1994).** "Business-Cycle Phases and Their Transitional Dynamics." *Journal of Business & Economic Statistics* 12(3), 299–308. [publisher — paywalled] — The applied template for time-varying transitions.
- **Grinsztajn, L., Oyallon, E. & Varoquaux, G. (2022).** "Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data?" NeurIPS Datasets and Benchmarks. — Why trees suit irregular targets and uninformative inputs.
- **Chen, L., Pelger, M. & Zhu, J. (2024).** "Deep Learning in Asset Pricing." *Management Science* 70(2), 714–750. — Learned macroeconomic states inside an asset-pricing objective.
- **Hinton, G., Vinyals, O. & Dean, J. (2015).** "Distilling the Knowledge in a Neural Network." arXiv:1503.02531. — Training a student on a teacher's soft outputs.
- **Frénay, B. & Verleysen, M. (2014).** "Classification in the Presence of Label Noise: A Survey." *IEEE Transactions on Neural Networks and Learning Systems* 25(5), 845–869. [publisher — paywalled] — Which kinds of label noise classifiers tolerate.
- **Adams, R. P. & MacKay, D. J. C. (2007).** "Bayesian Online Changepoint Detection." arXiv:0710.3742. — The run-length posterior.
- **Breiman, L. (2001).** "Random Forests." *Machine Learning* 45(1), 5–32. [publisher — paywalled] — Averaged trees grown on bootstrap samples.
- **Friedman, J. H. (2001).** "Greedy Function Approximation: A Gradient Boosting Machine." *Annals of Statistics* 29(5), 1189–1232. — Gradient boosting.
- **Koenker, R. & Bassett, G. (1978).** "Regression Quantiles." *Econometrica* 46(1), 33–50. [publisher — paywalled] — Quantile regression, the basis of growth-at-risk.
- **Page, E. S. (1954).** "Continuous Inspection Schemes." *Biometrika* 41(1–2), 100–115. [publisher — paywalled] — The CUSUM change detector.
- **Bourlard, H. & Morgan, N. (1994).** *Connectionist Speech Recognition: A Hybrid Approach.* Kluwer. [publisher — paywalled] — Neural classifiers inside a hidden Markov model, the template for §5.11.
- **Niculescu-Mizil, A. & Caruana, R. (2005).** "Predicting Good Probabilities with Supervised Learning." *Proceedings of the 22nd International Conference on Machine Learning*, 625–632. — Which classifiers need calibration, and how.
- **Cohen, J. (1960).** "A Coefficient of Agreement for Nominal Scales." *Educational and Psychological Measurement* 20(1), 37–46. [publisher — paywalled] — Kappa.

10.4 Evaluation

- **Brier, G. W. (1950).** "Verification of Forecasts Expressed in Terms of Probability." *Monthly Weather Review* 78(1), 1–3. — The Brier score.

- **Murphy, A. H. (1973).** “A New Vector Partition of the Probability Score.” *Journal of Applied Meteorology* 12(4), 595–600. — Reliability, resolution and uncertainty.
- **Gneiting, T. & Raftery, A. E. (2007).** “Strictly Proper Scoring Rules, Prediction, and Estimation.” *Journal of the American Statistical Association* 102(477), 359–378. [publisher — paywalled] — The theory of proper scores.
- **Saito, T. & Rehmsmeier, M. (2015).** “The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets.” *PLOS ONE* 10(3), e0118432. — Why rare crisis classes need precision and recall.
- **Fawcett, T. (2006).** “An Introduction to ROC Analysis.” *Pattern Recognition Letters* 27(8), 861–874. [publisher — paywalled] — The standard tutorial on ROC curves and AUC.
- **Fawcett, T. & Provost, F. (1999).** “Activity Monitoring: Noticing Interesting Changes in Behavior.” *Proceedings of the Fifth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 53–62. — Evaluating detectors of change by delay and false alarms.
- **Politis, D. N. & Romano, J. P. (1994).** “The Stationary Bootstrap.” *Journal of the American Statistical Association* 89(428), 1303–1313. [publisher — paywalled] — Resampling dependent data in blocks.
- **Bailey, D. H. & López de Prado, M. (2014).** “The Deflated Sharpe Ratio.” *Journal of Portfolio Management* 40(5), 94–107. — Adjusting for the number of trials.
- **Harvey, C. R., Liu, Y. & Zhu, H. (2016).** “…and the Cross-Section of Expected Returns.” *Review of Financial Studies* 29(1), 5–68. [publisher — paywalled] — Multiple testing in finance.

10.5 Critiques and cautions

- **Welch, I. & Goyal, A. (2008).** “A Comprehensive Look at the Empirical Performance of Equity Premium Prediction.” *Review of Financial Studies* 21(4), 1455–1508. — Most return predictors fail out of sample: the evidence behind “direction is not learnable”.
- **Campbell, J. Y. & Thompson, S. B. (2008).** “Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?” *Review of Financial Studies* 21(4), 1509–1531. — The reply to Welch and Goyal: with simple restrictions, out-of-sample gains are small but positive.
- **Rapach, D. E., Strauss, J. K. & Zhou, G. (2010).** “Out-of-Sample Equity Premium Prediction: Combination Forecasts and Links to the Real Economy.” *Review of Financial Studies* 23(2), 821–862. — Combining many weak predictors recovers a little out-of-sample skill.
- **Cederburg, S., O’Doherty, M. S., Wang, F. & Yan, X. S. (2020).** “On the Performance of Volatility-Managed Portfolios.” *Journal of Financial Economics* 138(1), 95–117. [publisher — paywalled] — Out of sample, volatility management rarely beats the unmanaged strategy.
- **Moreira, A. & Muir, T. (2017).** “Volatility-Managed Portfolios.” *Journal of Finance* 72(4), 1611–1644. — The case for volatility timing; read with Cederburg et al.
- **Brown, G. W. & Cliff, M. T. (2004).** “Investor Sentiment and the Near-Term Stock Market.” *Journal of Empirical Finance* 11(1), 1–27. [publisher — paywalled] — Sentiment and breadth move with returns but predict little.
- **Wang, Y.-H., Keswani, A. & Taylor, S. J. (2006).** “The Relationships between Sentiment, Returns and Volatility.” *International Journal of Forecasting* 22(1), 109–123. — Sentiment measures are caused by returns and volatility.
- **Hamilton, J. D. (2018).** “Why You Should Never Use the Hodrick-Prescott Filter.” *Review of Economics and Statistics* 100(5), 831–843. — Two-sided filters and their end-point revisions.
- **Forbes, K. J. & Rigobon, R. (2002).** “No Contagion, Only Interdependence: Measuring Stock Market Comovements.” *Journal of Finance* 57(5), 2223–2261. — Correlation rises mechanically with volatility.
- **Brown, S. J., Goetzmann, W. N., Ibbotson, R. G. & Ross, S. A. (1992).** “Survivorship Bias in Performance Studies.” *Review of Financial Studies* 5(4), 553–580. [publisher — paywalled] — How conditioning on survivors distorts history.

10.6 Books

- **Williams, D. (1991).** *Probability with Martingales.* Cambridge University Press. — Conditional expectation and filtrations, rigorously.
- **Hamilton, J. D. (1994).** *Time Series Analysis.* Princeton University Press. — Autoregressions, persistence and regime-switching models.
- **Hastie, T., Tibshirani, R. & Friedman, J. (2009).** *The Elements of Statistical Learning*, 2nd ed. Springer. — Logistic regression, shrinkage, clustering, trees and boosting; free from the authors.
- **Jolliffe, I. T. (2002).** *Principal Component Analysis*, 2nd ed. Springer. — The reference on principal components.

10.7 Practitioner sources

Practitioner frameworks are often the clearest statement of how regimes are used, and come from firms that sell products built on them.

- **Podolsky, P., Johnson, R. & Jennings, O. (2012).** "The All Weather Story." Bridgewater Associates. — Four growth–inflation environments and risk balance across them. The firm manages All Weather portfolios.
- **Greetham, T. and colleagues (2004).** *The Investment Clock.* Merrill Lynch. — Reflation, recovery, overheat and stagflation. No public copy located.
- **Faber, M. T. (2007).** "A Quantitative Approach to Tactical Asset Allocation." *Journal of Wealth Management* 9(4), 69–79. [publisher — paywalled] — The ten-month moving-average rule. The author's firm runs trend-following funds.

10.8 Data sources

- Kenneth R. French data library — daily US market and risk-free returns from 1926.
- Cboe VIX historical data — the VIX and its term-structure and related indices.
- FRED, Federal Reserve Bank of St. Louis — BAA10Y, T10Y3M, USREC, NFCI and STLFSI4; ALFRED for vintages.
- Chicago Fed NFCI — the index, its subindices and documentation.
- OFR Financial Stress Index — daily, with category and regional contributions.

10.9 If you only read eight things

1. **Ang & Timmermann (2012)** — the review of regime models in finance.
2. **Kole & van Dijk (2017)** — rules against models for bull and bear markets.
3. **Lunde & Timmermann (2004)** — what a turning-point label is and how it behaves.
4. **López de Prado (2018)** — labels with reach, and validation that respects it.
5. **Bekaert & Hoerova (2014)** — what the VIX contains.
6. **Adrian, Boyarchenko & Giannone (2019)** — what financial conditions forecast, and what they do not.
7. **Welch & Goyal (2008)** — why direction labels disappoint.
8. **Gneiting & Raftery (2007)** — how to score a probability.

Appendix A. Concepts and prerequisites

Everything the main text uses without stopping to explain. Entries are ordered by **dependency**, not alphabetically, so later entries build on earlier ones. Each gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper. Several entries summarise material that the companion chapter, Market Regimes and Hidden Markov Models, develops at length; they link to its appendix rather than repeat it.

Index. Where each concept is first used:

Concept	First used	Concept	First used
Information sets and look-ahead	§1.4	Clustering	§4.6
Conditional probability forecasts	§1.4	Trees and boosting	§5.5
Persistence and half-life	§1.4	Quantile regression	§5.10
Sampling error of correlations	§2.5	Hysteresis	§1.4
Proper scores and the Brier score	§1.4	Two-sided filters	§2.8
ROC curves and AUC	§1.4	Markov chains and durations	§5.11
Cohen's kappa	§1.1	Hidden Markov models and filtering	§1.5
Information coefficient and hit rate	§1.4	Statistical jump models	§4.6
Sharpe ratio and drawdown	§1.1	Change-point detection	§5.9
Logistic regression	§5.4	Knowledge distillation	§4.10

Concept	First used	Concept	First used
Shrinkage	§5.4	Realised volatility	§3.2
Calibration	§5.11	Implied volatility and the VIX	§2.3
Overlapping labels	§4.5	Credit spreads	§3.4
Censoring and provisional labels	§4.1	The term spread	§2.8
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A.1 Information sets and look-ahead

The idea. A backtest is honest only if every number it uses was available at the moment the decision was made. The *information set* $\mathcal{F}_t$ is the bookkeeping for that: everything published by the close of day t, in the version published then. Look-ahead bias is the use of anything outside it.

Formally. $\{\mathcal{F}_t\}$ is a filtration: an increasing family of information sets, $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$. A quantity is $\mathcal{F}_t$-measurable if its value is determined by $\mathcal{F}_t$. In practice the test is operational: could this number have been computed from data published by the close of day t, in the version that existed then? A revised series has a *vintage* for each publication date, and only the vintage current at t belongs to $\mathcal{F}_t$.

Why it appears here. The reach of a label (§1.4) says which information set the label belongs to. The timing table of §3.11 lists when each input enters $\mathcal{F}_t$, and its rules keep features $\mathcal{F}_t$-measurable.

Deeper. Appendix A.1 of Market Regimes and Hidden Markov Models; Williams (1991), chapters 9 and 10; on vintages, Croushore & Stark (2001).

A.2 Conditional probability forecasts

The idea. For a binary label, the best possible forecast is not a guess of 0 or 1 but the probability of 1 given everything known. Every classifier is an attempt to estimate that probability.

Formally. $p_t = \Pr(y_t = 1 \mid \mathcal{F}_t) = \mathbb{E}[y_t \mid \mathcal{F}_t]$. Among all forecasts built from $\mathcal{F}_t$ it minimises the expected squared error, and the minimum is $\mathbb{E}[\mathrm{Var}(y_t \mid \mathcal{F}_t)] = \mathbb{E}[p_t(1-p_t)]$, because a binary variable with success probability p has variance $p(1-p)$.

Why it appears here. It is the ceiling on any classifier's accuracy (§1.4), and the reason a label with zero reach needs no model: then p_t is 0 or 1 and the ceiling is perfection.

Deeper. Williams (1991), chapter 9, on conditional expectation as the best mean-squared-error predictor.

A.3 Persistence and half-life

The idea. A persistent series stays near where it is. Persistence is why yesterday's regime predicts today's, why volatility can be forecast, and why daily samples contain far fewer independent observations than days.

Formally. For a first-order autoregression $x_t = \mu + \rho\,(x_{t-1} - \mu) + e_t$ with $0 < \rho < 1$, a deviation from the mean shrinks by the factor ρ each period, so after k periods it is ρ^k of its size and it halves after $\ln 0.5 / \ln \rho$ periods. With $\rho = 0.98$ per day the half-life is about 34 days. A slope of 0.72 per month gives $\ln 0.5 / \ln 0.72 \approx 2.1$ months.

Why it appears here. The persistence of volatility makes volatility labels learnable and the stale label a strong baseline (§1.4, §5.1). The lack of persistence in returns makes direction labels flicker (§4.5).

Deeper. Hamilton (1994), chapter 3.

A.4 Sampling error of correlations

The idea. A correlation or autocorrelation estimated from a short window is noisy, and the noise can be as large as the variation the estimate is meant to detect.

Formally. For n independent pairs with true correlation ρ, the estimate's standard error is approximately $(1-\rho^2)/\sqrt{n}$, which is $1/\sqrt{n}$ at $\rho = 0$: about 0.13 for $n = 63$. The approximation needs n moderately large and the pairs independent; overlapping or persistent data make the error larger. Fisher's transformation $\frac{1}{2}\ln[(1+r)/(1-r)]$ gives an approximately normal statistic with standard error $1/\sqrt{n-3}$, which is the better basis for intervals.

Why it appears here. It limits trending-versus-mean-reverting regimes (§2.5) and correlation regimes (§2.9, §3.8).

Deeper. Any mathematical statistics text on the sampling distribution of the correlation coefficient.

A.5 Proper scores and the Brier score

The idea. A scoring rule grades a probability forecast against what happened. It is *proper* if no forecaster can improve their expected score by reporting anything other than their true belief, which makes scores comparable and honest.

Formally. The Brier score is $\frac{1}{n}\sum_t(\hat{p}_t - y_t)^2$ and the log loss $-\frac{1}{n}\sum_t[y_t \ln \hat{p}_t + (1-y_t)\ln(1-\hat{p}_t)]$. Both are strictly proper: for an event with true probability p, the expected Brier score of a forecast q is $p(1-q)^2 + (1-p)q^2$, which is minimised at $q = p$. Skill against a reference forecast is $1 - \mathrm{BS}/\mathrm{BS}_{\mathrm{ref}}$. Murphy's decomposition writes the Brier score as reliability minus resolution plus uncertainty (§7.2).

Why it appears here. Every Brier skill score in §5–§7 is this quantity, with climatology or the stale label as the reference.

Deeper. Gneiting & Raftery (2007); Murphy (1973).

A.6 ROC curves and AUC

The idea. A classifier's score can be turned into yes-or-no decisions at any threshold. The receiver operating characteristic (ROC) curve shows, for every threshold, the share of positives caught against the share of negatives falsely flagged. The area under it summarises ranking quality in one number.

Formally. With true-positive rate $\mathrm{TPR}(c) = \Pr(\hat{p} > c \mid y = 1)$ and false-positive rate $\mathrm{FPR}(c) = \Pr(\hat{p} > c \mid y = 0)$, the ROC curve plots TPR against FPR as the threshold c varies, and $\mathrm{AUC} = \Pr(\hat{p}_i > \hat{p}_j \mid y_i = 1, y_j = 0)$, with ties counted as one half. This is the Mann–Whitney statistic, so the AUC is unchanged by any increasing transformation of the scores. If the scores of positives and negatives are normal with equal variance and means d' standard deviations apart, $\mathrm{AUC} = \Phi(d'/\sqrt{2})$, where Φ is the standard normal distribution function. The precision–recall curve plots $\Pr(y = 1 \mid \hat{p} > c)$ against $\mathrm{TPR}(c)$ and, unlike the ROC curve, depends on the base rate.

Why it appears here. AUC is the ranking score of §1.4, §6.4 and §7.3. It is blind to calibration, which is why the chapter pairs it with the Brier score.

Deeper. Fawcett (2006); Saito & Rehmsmeier (2015).

A.7 Cohen's kappa

The idea. Two labels can agree on most days simply because both say "calm" most of the time. Kappa corrects for that, measuring agreement beyond what the two base rates would produce by chance.

Formally. With p_o the observed share of days on which two binary labels agree, and $p_e = \pi_1\pi_2 + (1-\pi_1)(1-\pi_2)$ the agreement expected if they were independent with base rates π_1 and π_2, $\kappa = (p_o - p_e)/(1-p_e)$. It is 1 for perfect agreement and 0 for chance-level agreement, and it can be negative. Its maximum attainable value falls when the two base rates differ.

Why it appears here. The agreement table of §6.2 and the stability check of §4.9.

Deeper. Cohen (1960).

A.8 Information coefficient and hit rate

The idea. A forecast of returns is judged by how well it ranks or correlates with what happens. Even a good forecast is right only slightly more often than a coin.

Formally. The information coefficient is the correlation between forecast and outcome. If both are jointly normal with correlation IC, the probability that they share a sign, the hit rate, is $\frac{1}{2} + \arcsin(\mathrm{IC})/\pi$: 51.6% at an IC of 0.05.

Why it appears here. It calibrates what a direction label can offer (§1.4).

Deeper. Section 2 of Edge as Information.

A.9 Sharpe ratio and drawdown

The idea. The Sharpe ratio measures return per unit of risk; the drawdown measures the worst fall from a peak. One describes the average experience, the other the worst.

Formally. The annualised Sharpe ratio of daily excess returns e_t is $\sqrt{A}\,\bar{e}/s_e$, with $\bar{e}$ and s_e their mean and standard deviation. The drawdown at t is $1 - W_t/\max_{s \le t} W_s$ for a wealth path W, and the maximum drawdown is its largest value over the sample.

Why it appears here. The value of the bull–bear label, known in advance or when confirmed, is measured by both (§1.1, §6.3).

Deeper. Section 4 of Edge as Information; Appendix A.25 of the companion chapter, on the Sharpe ratio's standard error.

A.10 Logistic regression

The idea. Model the log-odds of an event as a weighted sum of inputs. The model produces probabilities between 0 and 1, and its coefficients say how much each input moves the odds.

Formally. $\Pr(y_t = 1 \mid x_t) = \Lambda(\beta_0 + \beta^\top x_t)$ with $\Lambda(u) = 1/(1+e^{-u})$, so $\ln[p/(1-p)] = \beta_0 + \beta^\top x_t$. The coefficients maximise the log-likelihood $\sum_t [y_t \ln \hat{p}_t + (1-y_t)\ln(1-\hat{p}_t)]$, which is concave, so Newton's method (iteratively reweighted least squares) finds the unique maximum in a few steps. The probit model replaces Λ with Φ; the ordered logit applies one coefficient vector to $K-1$ cumulative splits of ordered classes; dynamic versions add lagged labels or a lagged index as inputs.

Why it appears here. It is the default classifier of §5.4 and the model of the worked example (§6).

Deeper. Hastie, Tibshirani & Friedman (2009), section 4.4.

A.11 Shrinkage

The idea. With few independent observations, estimated coefficients are noisy, and a model that uses them at face value fits noise. Shrinkage pulls the coefficients toward zero, trading a little bias for much less variance.

Formally. Ridge regularisation adds $\frac{1}{2}\,\alpha\, n \|\beta\|^2$ to the negative log-likelihood of n rows (§5.4); the lasso uses a multiple of $\|\beta\|_1$ instead and sets some coefficients exactly to zero. As the penalty grows the fitted probabilities move toward the base rate. The penalty is chosen by validation or, as in the worked example, fixed in advance.

Why it appears here. In the worked example the twelve-input model lost to a one-input model without shrinkage and matched it with shrinkage (§5.4, §6.4), because the effective sample is a few dozen episodes.

Deeper. Hastie, Tibshirani & Friedman (2009), section 3.4; Appendix A.31 of the companion chapter, on shrinkage as partial pooling.

A.12 Calibration

The idea. A score that ranks days well need not be a probability. Calibration maps scores to probabilities that match observed frequencies.

Formally. Platt scaling fits $\Pr(y = 1 \mid s) = \Lambda(a + b\,s)$ to a score s by logistic regression; isotonic regression fits the best non-decreasing step function of the score. Both must be fitted on data the scoring model did not train on, out-of-fold or on a later period, because a flexible model's scores on its own training rows are overconfident.

Why it appears here. Post-processing (§5.11) and the reliability check of §7.2; tree ensembles need it most (§5.5).

Deeper. Niculescu-Mizil & Caruana (2005).

A.13 Overlapping labels

The idea. A label defined over the next 63 days shares 62 days of outcome with the label of the following day. A training set of thousands of daily rows then holds far fewer independent pieces of evidence than it appears to.

Formally. For labels over windows of length h, rows at least h days apart share no outcome days, so n days contain roughly n/h non-overlapping labels. When the label is also persistent beyond its window the effective count is lower still, and for a rare regime it approaches the number of episodes. López de Prado's *average uniqueness* weights each row by the inverse of the number of labels its window overlaps.

Why it appears here. It is why §4.10 purges, weights and sizes test sets by episodes, and why the worked example's intervals resample years (§6.1).

Deeper. López de Prado (2018), chapter 4; Appendix A.30 of the companion chapter, on effective sample size.

A.14 Censoring and provisional labels

The idea. At the end of any sample, some labels are not yet decided: the forward window has not elapsed, or the current phase has not ended. Treating the undecided labels as decided is a look-ahead in reverse: it assumes an outcome that has not happened.

Formally. A label with fixed reach b is *censored* for the last b days of the sample. A turning-point label is final up to the current running extreme and provisional after it (§4.10). Survival analysis treats the same problem for durations: a spell still in progress has a known lower bound on its length but no end.

Why it appears here. §4.1, §4.4 and the training rule of §4.10.

Deeper. Any survival-analysis text on right-censoring.

A.15 Purging, embargo and walk-forward validation

The idea. In time-series validation, information leaks from test to training through overlapping labels and through serial dependence. Purging removes the leaking rows; an embargo adds a gap; walk-forward testing only ever predicts the future from the past.

Formally. Walk-forward validation refits at dates $\tau_1 < \tau_2 < \dots$ on rows before each τ_k and tests on $[\tau_k, \tau_{k+1})$. Purging drops any training row whose label window overlaps the test period; an embargo also drops training rows within a fixed gap after it. For a label with reach b, the purge at a refit date τ removes training rows with $t > \tau - b$.

Why it appears here. §4.10 and the worked example's training rule ("rows whose label was final by the refit date").

Deeper. López de Prado (2018), chapters 7 and 12; section 11 of the companion chapter and its Appendix A.39.

A.16 Block bootstrap and permutation nulls

The idea. The ordinary bootstrap resamples days independently and so destroys the persistence that makes neighbouring days alike. Resampling blocks of consecutive days keeps it, and gives honest error bars for dependent data. A permutation null asks what score the same pipeline would get if features and labels were unrelated but each kept its own persistence.

Formally. The moving-block bootstrap draws blocks of fixed length with replacement and concatenates them; the stationary bootstrap draws blocks of random, geometrically distributed length (Politis & Romano, 1994). A persistence-preserving permutation shifts the label series circularly against the features by a random offset, refits and records the score; repeated, it gives the score's distribution under no relationship.

Why it appears here. The 90% intervals of §6.4 resample one-year blocks; §7.6 recommends both tools.

Deeper. Politis & Romano (1994); Appendix A.41 of the companion chapter.

A.17 Multiple testing

The idea. Try enough label definitions, thresholds and inputs and one of them will look good by luck. The best of many trials overstates what any one of them will deliver.

Formally. If N independent strategies with no edge are tested, the expected best Sharpe ratio grows roughly with $\sqrt{2 \ln N}$ times its standard error. Corrections either raise the significance threshold with the number of trials or deflate the best result for them.

Why it appears here. §3.1's warning against testing forty inputs on thirty episodes, and the trial count of §7.6.

Deeper. Harvey, Liu & Zhu (2016); Bailey & López de Prado (2014); Appendix A.40 of the companion chapter.

A.18 Principal components and factor models

The idea. When many inputs move together, a few combinations of them carry most of the variation. Principal components find those combinations; a factor model treats them as unobserved drivers; the Mahalanobis distance measures how unusual a day is once the inputs' co-movement is taken into account.

Formally. For standardised inputs with correlation matrix C, the principal components are the eigenvectors of C, and the share of total variance explained by the k-th is its eigenvalue divided by the sum of all eigenvalues. A dynamic factor model writes each input as loadings times a few latent factors that follow their own time-series process. The squared Mahalanobis distance of a vector R from mean $\bar{R}$ with covariance Σ is $(R-\bar{R})^{\top}\Sigma^{-1}(R-\bar{R})$.

Why it appears here. The STLFSI4 is a first principal component and the NFCI comes from a dynamic factor model (§2.7); turbulence is a Mahalanobis distance and the absorption ratio a principal-component share (§2.7, §5.3); §3.13 measures the input space with them.

Deeper. Jolliffe (2002); Appendix A.23 of the companion chapter.

A.19 Clustering

The idea. Clustering groups days whose features look alike, without labels. Each group can then be read as a regime.

Formally. k-means chooses K centroids and assigns each day to the nearest, minimising the total squared distance. A Gaussian mixture gives each cluster its own mean and covariance and assigns days probabilistically, fitted by the EM algorithm. Clustering whole return distributions instead of feature vectors needs a distance between distributions, such as the Wasserstein distance (Horvath, Issa & Muguruza, 2024). None of these methods knows that days are ordered in time, so their regimes flicker unless persistence is added (A.26).

Why it appears here. Model-derived labels (§4.6) and the Gaussian-mixture regimes of §2.8.

Deeper. Hastie, Tibshirani & Friedman (2009), sections 8.5 and 14.3; Appendix A.22 of the companion chapter.

A.20 Trees and boosting

The idea. A decision tree splits the input space by thresholds and predicts a constant in each region. Many trees combined, by averaging or by adding them in sequence, make a flexible model of interactions and non-linear effects.

Formally. A random forest averages deep trees grown on bootstrap samples with random subsets of inputs (Breiman, 2001). Gradient boosting adds shallow trees one at a time, each fitted to the gradient of the loss at the current prediction (Friedman, 2001). Because each tree predicts a constant beyond the last split, the ensemble cannot extrapolate past the range of its training data.

Why it appears here. §5.5, and the student model of the teacher–student pipelines in §5.8.

Deeper. Hastie, Tibshirani & Friedman (2009), chapters 10 and 15; Appendix A.33 of the companion chapter.

A.21 Quantile regression

The idea. Ordinary regression models the average outcome. Quantile regression models a chosen quantile, such as the worst 5% of outcomes, so it can describe how the lower tail moves with conditions even when the average does not.

Formally. The q-th quantile regression minimises $\sum_t \rho_q(y_t - x_t^{\top}\beta)$ with the check loss $\rho_q(u) = u\,(q - \mathbb{1}[u<0])$, which weights positive errors by q and negative errors by $1-q$.

Why it appears here. Growth-at-risk (§2.7) and the direct forecast of drawdown risk in §5.10.

Deeper. Koenker & Bassett (1978).

A.22 Hysteresis

The idea. A switch with a single threshold chatters when its input hovers near the threshold. Hysteresis gives it memory: a higher threshold to switch on, a lower one to switch off.

Formally. With entry threshold u_{in} above exit threshold u_{out}, the state turns on when the input reaches u_{in}, turns off when it falls below u_{out}, and otherwise stays as it was (§5.2). The gap $u_{\text{in}} - u_{\text{out}}$ trades delay against the number of switches.

Why it appears here. The real-time ±20% rule is a hysteresis rule (§1.4), and hysteresis is the cheapest cure for flickering regimes (§4.2, §5.11).

Deeper. Section 12 of the companion chapter, on hysteresis bands and trading costs.

A.23 Two-sided filters

The idea. Many smoothing methods estimate the trend at a date from data on both sides of it. They look clean in history but cannot be computed in real time, and their recent values change as new data arrive.

Formally. A two-sided filter computes $\tilde{x}_t = \sum_{k=-m}^{m} w_k x_{t+k}$, using m future values. At the end of a sample it must fall back on a one-sided version, so its last values are revised as data accumulate. The Hodrick–Prescott filter chooses the trend τ_t to minimise $\sum_t (x_t - \tau_t)^2 + \lambda_{\text{HP}} \sum_t (\Delta^2 \tau_t)^2$ over the whole sample, so every fitted value depends on every observation.

Why it appears here. Labels defined by two-sided filters share the look-ahead of turning-point labels (§4.4), and the OECD's leading indicators use Hodrick–Prescott detrending (§2.8).

Deeper. Hamilton (2018).

A.24 Markov chains and durations

The idea. A Markov chain moves between a few states with fixed probabilities that depend only on the current state. The probability of staying put sets how long states last.

Formally. With transition probabilities $p_{jk} = \Pr(s_{t+1} = k \mid s_t = j)$, a visit to state j lasts a geometric number of days with mean $1/(1 - p_{jj})$: a daily stay probability of 0.98 gives visits averaging 50 days. The stationary distribution gives the long-run share of time in each state.

Why it appears here. The Markov filter that smooths a classifier's output (§5.11) and the latent-state models of §5.7.

Deeper. Appendices A.4 and A.6 of the companion chapter.

A.25 Hidden Markov models and filtering

The idea. A hidden Markov model treats the regime as an unobserved Markov state and infers it from data. Inference comes in three forms that use different information: *predicted* (data up to yesterday), *filtered* (up to today) and *smoothed* (the whole sample).

Formally. The filter alternates two steps: carry yesterday's belief through the transition matrix, then reweight each state by how well it explains today's observation and renormalise (the Hamilton filter). The smoother adds a backward pass that uses later data (the Kim smoother). Dividing a classifier's probability by its training prior gives a quantity proportional to the likelihood of the observation in each state, which can take the place of the reweighting step (§5.11).

Why it appears here. The dating, nowcasting and forecasting tasks of §1.5 are the smoothed, filtered and predicted beliefs; smoothed probabilities used as labels (§4.6) carry future data.

Deeper. Sections 5 and 6 and Appendices A.10–A.11 of the companion chapter; Bourlard & Morgan (1994) for the hybrid with a classifier.

A.26 Statistical jump models

The idea. A jump model assigns each day to the state whose typical features it is closest to, as clustering does, but charges a fixed penalty for every switch, so states persist.

Formally. It chooses states $s_1, \ldots, s_T$ and centroids $\mu_1, \ldots, \mu_K$ to minimise $\sum_t \|x_t - \mu_{s_t}\|^2 + \lambda \sum_t \mathbb{1}[s_t \neq s_{t-1}]$. Given the centroids, the best state sequence is found exactly by dynamic programming, the same recursion as the Viterbi algorithm

for hidden Markov models; given the states, each centroid is the mean of its days. Running the recursion forward only and reading off the best state at each date gives the causal version.

Why it appears here. Model-derived labels (§4.6), teacher–student pipelines (§5.8) and the worked example's jump-model label (§6).

Deeper. Bemporad et al. (2018); Nystrup, Lindström & Madsen (2020); section 5 of the companion chapter.

A.27 Change-point detection

The idea. Instead of asking which of a few known states the market is in, ask whether its behaviour has just changed.

Formally. A cumulative-sum (CUSUM) detector adds up evidence for a shift, $C_t = \max(0, C_{t-1} + \ell_t)$ with ℓ_t a log-likelihood ratio of "changed" against "unchanged", and alarms when C_t crosses a threshold (Page, 1954). Bayesian online change-point detection keeps a posterior over the run length, the time since the last change, and updates it with each observation (Adams & MacKay, 2007).

Why it appears here. §5.9, as a monitoring layer beside a classifier.

Deeper. Appendix A.21 of the companion chapter.

A.28 Knowledge distillation

The idea. A large or awkward model, the teacher, can train a simpler one, the student, by letting the student imitate the teacher's outputs. Imitating the teacher's probabilities passes on more than imitating its hard decisions, because the probabilities carry the teacher's uncertainty.

Formally. The student minimises a loss against the teacher's soft targets, for example the cross-entropy $-\sum_t \sum_k q_t^{(k)} \ln \hat{p}_t^{(k)}$ with $q_t^{(k)}$ the teacher's probability of class k, instead of against one-hot labels.

Why it appears here. Teacher–student regime pipelines (§5.8) are distillation with a two-sided teacher, and soft labels (§4.10) are its targets.

Deeper. Hinton, Vinyals & Dean (2015).

A.29 Realised volatility

The idea. Volatility is not observed directly, but the sum of squared returns over a window estimates it, and the more finely the window is sampled the better the estimate.

Formally. $\mathrm{RV}_t^{(n)} = \sqrt{(A/n)\sum_{i=0}^{n-1} \ell_{t-i}^2}$ annualises the average squared daily log return over n days. With intraday data, summing squared five-minute returns gives a far more precise daily variance. The heterogeneous autoregressive (HAR) model forecasts realised variance from its daily, weekly and monthly averages.

Why it appears here. It is the main price-path input (§3.2), the stale label for volatility targets (§1.4), and the basis of the forward-volatility label (§4.5).

Deeper. Andersen, Bollerslev, Diebold & Labys (2003); Corsi (2009); Appendix A.18 of the companion chapter.

A.30 Implied volatility, the VIX and the variance risk premium

The idea. Option prices imply how much volatility buyers expect and how much they will pay for protection against it. The VIX reads that from S&P 500 options for the next 30 days. Because option buyers pay for insurance, implied variance is usually above the variance then realised; the gap is the variance risk premium.

Formally. $\mathrm{VIX}_t^2/10^4$ approximates the expectation of the annualised variance of the S&P 500 over the next 30 days under *risk-neutral* probabilities, the probabilities implied by option prices, computed from a strip of out-of-the-money options. The variance risk premium is that expectation minus the expectation under real-world probabilities; a common proxy subtracts trailing realised variance. The VIX *term structure* compares the same measure over 9 days, 30 days, 3 months and 6 months. It normally slopes upward (contango) and inverts (backwardation) in stress.

Why it appears here. The VIX is the leading forward-looking input (§3.3), and the volatility regimes of §2.3 are bands on it.

Deeper. Sections 5 and 6 of Implied Volatility; the Cboe VIX methodology; Bekaert & Hoerova (2014).

A.31 Credit spreads and the excess bond premium

The idea. A corporate bond yields more than a government bond of similar maturity because it might default, because investors demand a premium for bearing that risk, and because it is harder to trade. The extra yield, the spread, rises with all three.

Formally. A yield spread subtracts a government yield from a corporate yield; the Baa–10-year spread uses Moody's Baa index and the 10-year Treasury. An option-adjusted spread is the constant spread over the government curve that prices a bond after removing the value of options embedded in it, such as the issuer's right to call it. The excess bond premium is the part of corporate spreads left after accounting for each firm's expected default, so it isolates the price of risk.

Why it appears here. Credit is the price-of-risk input of §3.4 and the second factor of §3.13.

Deeper. Sections 8 and 9 of Bonds and Bond Markets; Gilchrist & Zakrajšek (2012).

A.32 The term spread

The idea. Long-term interest rates embed expected future short rates. When markets expect the central bank to cut, typically because they expect a slowdown, long rates fall below short rates and the curve inverts.

Formally. The term spread is a long yield minus a short yield, for example the 10-year minus 3-month Treasury yield. The near-term forward spread compares the forward rate implied for six quarters ahead with the current short rate. Both also contain a term premium, the extra return demanded for holding long bonds, which varies over time.

Why it appears here. It is the classic recession predictor (§2.8, §3.5) and the slow second factor of §3.13.

Deeper. Section 11 of Bonds and Bond Markets; Estrella & Mishkin (1998).

A.33 Stress and conditions indices

The idea. Financial stress shows up at once in many places: spreads widen, volatility rises, funding markets strain. A stress index averages many such measurements so that the common movement stands out from the noise of each.

Formally. Each input is standardised against its history, and the index is a weighted sum, $I_t = \sum_j w_j \tilde{x}_{j,t}$, with weights from a first principal component, a dynamic factor model, or a correlation-based scheme (§5.3). Indices are usually scaled so that zero is the historical average.

Why it appears here. §2.7 uses them as regimes and §3.6 as inputs; their revisions and full-sample weights are the main hazards.

Deeper. Kliesen & Smith (2010); Brave & Butters (2012); Hollo, Kremer & Lo Duca (2012); Monin (2019).

A.34 Survivorship bias

The idea. A data set that contains only the survivors, such as today's index members or funds still in business, describes a past better than the one that happened, because the losers have been removed.

Formally. If membership at the end of the sample depends on performance during it, any statistic computed over today's members is conditioned on an outcome that was not known at the time, which is a form of look-ahead.

Why it appears here. Breadth and correlation inputs built from current index members (§3.7, §3.8).

Deeper. Brown, Goetzmann, Ibbotson & Ross (1992).

A.35 Variance ratio and Hurst exponent

The idea. If returns are independent, the variance of q-day returns is q times the variance of one-day returns. Departures from that scaling reveal trending or mean reversion.

Formally. $\mathrm{VR}(q) = \mathrm{Var}(r_{t-q+1} + \cdots + r_t)/[q\,\mathrm{Var}(r_t)]$, which equals $1 + 2\sum_{k=1}^{q-1}(1 - k/q)\rho_k$ in terms of the return autocorrelations ρ_k: above one when they are positive (trending), below one when negative (mean reverting). The Hurst exponent H describes the same scaling as a power law, with the range of cumulative deviations growing like n^H; $H = \frac{1}{2}$ for independent returns.

Why it appears here. The trending-versus-mean-reverting regimes of §2.5.

Deeper. Lo & MacKinlay (1988).

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